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Brain-Computer Interface for Prosthetic Device Using EMG

A thesis submitted for the degree of Bachelor of Science in Mechatronics Engineering

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Year
2025

This is to certify that:

- (i) the thesis comprises only my original work towards the Bachelor Degree
- (ii) due acknowledgement has been made in the text to all other material used

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Date: 29/5/2025

Abstract

This thesis presents the design, development, and validation of a low-cost, tendon-driven bionic hand controlled by surface electromyography signals for below-elbow wrist amputees. The thesis aims at creating a prosthetic hand that can address challenges faced by amputees in developing countries like Egypt, mainly addressing cost and functionality. The prosthetic employs a hybrid actuation system combining servo and geared direct current motors to achieve independent finger control and mimic human movements. Inspired by human biomechanics, the bionic uses tendon driven actuation, utilizing fishing braid lines and elastic bands for smooth flexion and passive extension, with damping rubber between each joint to simulate cartilages. An adaptive control algorithm with dynamic thresholding processes the surface electromyography signals from forearm muscles creating a brain computer interface and providing a binary output for motor control and enabling intuitive, real-time control without complex machine learning. The prototype was fabricated using 3D-printed Polylactic Acid and powered by a custom 2S LiPo batteries with a full USB-C charging system. Testing for functionality, revealed a 5 kg load-carrying capacity without holding torque, 100% accuracy in detecting grip and wrist flexion and 98.25% accuracy in detecting individual finger flexion, all while limiting the prosthetics' cost to 5000 EGP. This work establishes a scalable design for affordable, user-friendly prosthetics, with potential for future enhancements.

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List of Terms

ADC	Analog to Digital Converter
CNC	Computer Numerical Control
DIP	Distal Interphalangeal Joint
DOF	Degrees of Freedom
EMA	Exponential Moving Average
EMG	Electromyography
FDM	Fused Deposition Modeling
IP	Interphalangeal Joint
MA	Moving Average
MCP	Metacarpophalangeal Joint
PIP	Proximal Interphalangeal Joint
PLA	Polylactic Acid
TPU	Thermoplastic Polyurethane

List of Symbols

S: Shortest link length in Grashof's Law (mm)

L: Longest link length in Grashof's Law (mm)

P, Q: Other link lengths in Grashof's Law (mm)

r: Length of link in loop-closure equation (mm)

θ : Counterclockwise angle between link and x-axis (degrees)

RPM: Revolutions per minute of motor

t: Time for finger flexion (seconds)

s: Distance traveled by wire for full flexion (mm)

T: Torque of the motor (N·cm)

F: Maximum force finger can withstand (N)

$V_{smoothed(n)}$: Smoothed voltage value at step n (V)

α : Exponential Moving Average factor

$V_{current}$: Current voltage value captured by ADC (V)

$V_{smoothed(n-1)}$: Previous smoothed voltage value (V)

σ^2 : Variance of voltage samples

N: Number of samples

V_i : Voltage value inside buffer (V)

$V_{baseline}$: Baseline voltage (V)

$V_{threshold}$: Threshold voltage for muscle state detection (V)

K: Threshold multiplier

V_{on} : Minimum voltage for “on” state (V)

V_{off} : Maximum voltage for “off” state (V)

H: Hysteresis percentage

Chapter 1

Introduction

The loss of a limb, particularly a hand, imposes significant limitations on a person's ability to interact with the surrounding world. The highly dexterous structure of the human hand allows it to perform extremely complex tasks on a daily basis, which can appear very simple such as holding a pencil or using utensils. The precision, strength and coordination of the hand is often taken for granted until it is lost due to traumatic, pathologic or congenital causes. As of 2017, 57.7 million people were living with limb amputations due to traumatic causes worldwide [1].

Beyond physical limitations, the loss of a limb can severely impact mental health, body image, and social participation. The loss of independence alone can have a huge psychological toll, especially for people who recently experienced amputation, where they find difficulty navigating simple day to day tasks. Many individuals face depression, anxiety, and reduced quality of life due to the sudden change in their physical capabilities.

Historically, prosthetics have offered functional and cosmetic replacements aimed at addressing the aforementioned physical challenges as well as the societal and emotional complications faced by individuals with limb loss. Cosmetic prosthetics can be seen throughout history, with the first recorded appearance between 2,600 and 3,400. A toe known as the “Greville Chester toe” found in Egypt made from papier-mâché made from glue, linen, and plaster. [2] [figure 1.1]



Figure 1.1: Chester Toe

Years later functional prosthetics were created to provide the user with more utility. The most common forms of actuation for prosthetics found are operated through body powered mechanisms, mechanical linkages and tendons. Body powered mechanisms offer the simplest designs, but they provide the user with very restricted grip types, while linkage and tendon driven prosthetics offer a more dynamic range of motion and fine motor control with tendon driven mechanisms presenting more human-like movements

The incorporation of electronics and sensors led to huge advancements in bionics, especially electromyography (EMG) sensors that can detect muscle residual nerve signals but unfortunately its use is often restricted by high cost and complexity of training required for each person to enable effective use.

The journey from ancient innovation to modern engineering, however, spans centuries of evolving needs, materials, and understanding. In the 16th century, Ambroise Paré, a French military surgeon, introduced mechanical limbs with hinges and lockable joints, designed primarily for wounded soldiers returning from battle. His contributions laid the foundation for more functional, customizable devices [3] [Figure 1.2]

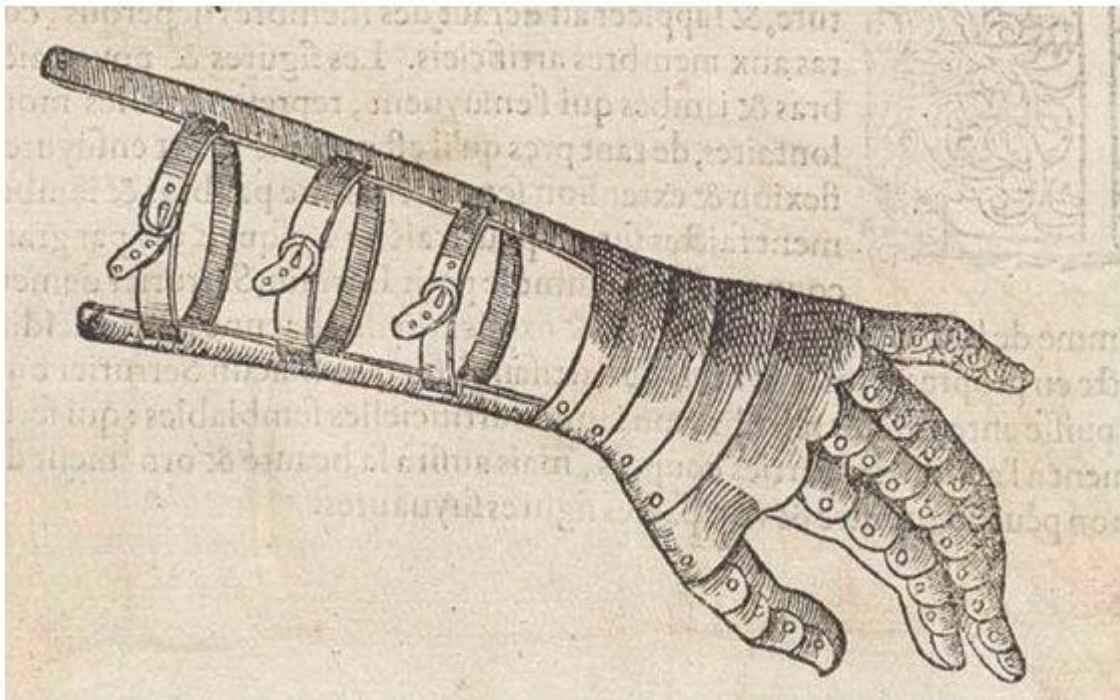


Figure 1.2: Drawing from barber/surgeon Ambroise Paré in the 1500s

By the 19th century, prosthetics had become increasingly sophisticated. The use of iron, leather, and springs allowed artisans to craft limbs that not only restored basic function but also reflected the era's growing focus on mechanical ingenuity and aesthetics. These devices were often designed for veterans of war. World War 2 sparked more research and early concepts of electromyography sensor controlled bionics were introduced [4].

Simultaneously, societal perceptions of limb loss began to shift. In Victorian England, prosthetic devices were not just medical aids but also cultural symbols. They appeared in literature and public life as representations of strength, resilience, or sometimes concealment, reflecting how society viewed disability at the time [5].

Despite the visible advancements in material and design, it wasn't until the 20th century that prosthetic technology truly began to incorporate biomedical engineering. From mechanical hooks and pulley systems, the field moved toward myoelectric prostheses devices capable of translating electrical muscle signals into controlled motion. This leap forward opened the door for what we now refer to as bionic limbs, merging engineering with biology to restore a semblance of natural hand function.

An example of modern prosthetics aspiring to combine both the functional and the aesthetic appeal, is Open Bionics' Deus Ex hand that was designed based on a futuristic video game and received praise for the sleek modern design. Leaning towards the modern design instead of a more human looking hand, led to better interactions where the narrative changed from empathetic to amazement [6] [figure 1.3].

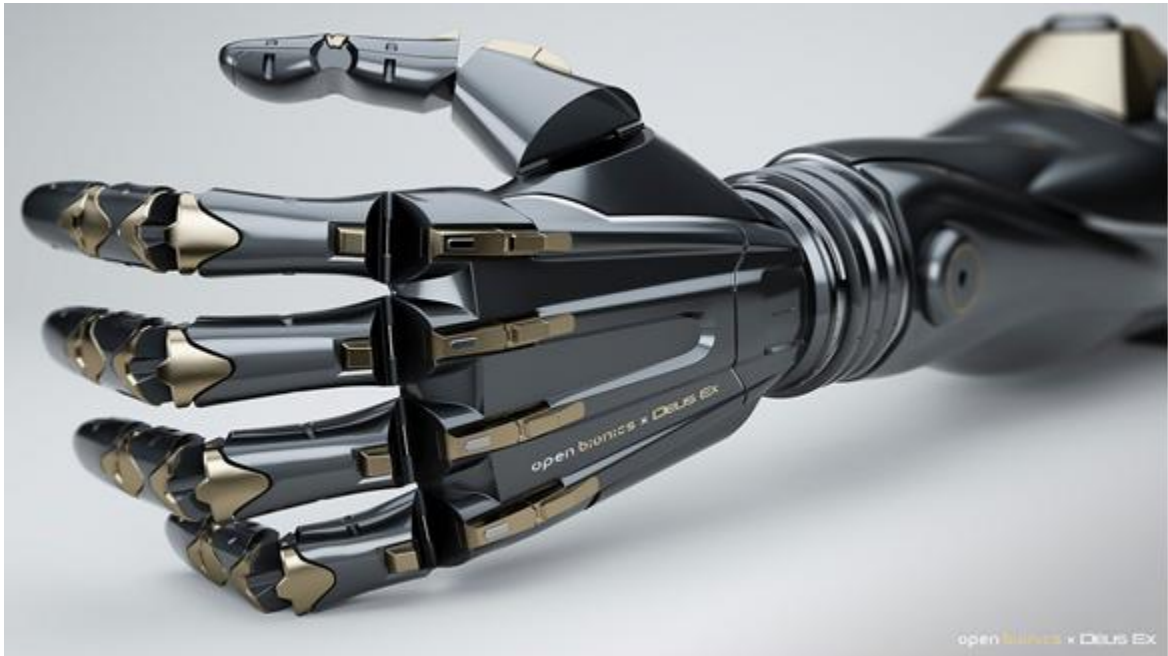


Figure 1.3: Deus Ex prosthetic

In addition to functional and cosmetic applications, prosthetics have also seen major development in the field of sports. Athletes with limb loss now have access to highly specialized prosthetic limbs designed to optimize performance and reduce the limitations that come with conventional devices with sports-specific prosthetics, such as carbon fiber running blades, are lightweight, energy-efficient, and built to replicate or even enhance the natural biomechanics of human limbs [7]. However, the use of advanced prosthetics in sports has also raised debates about fairness and whether certain devices give users an unfair advantage over non-amputee athletes, especially in high-profile cases like that of Oscar Pistorius [8]. Despite the controversy, the continuous evolution of prosthetic technology has allowed many amputees to not only return to sports but to also compete at elite levels, often with customized solutions designed for sprinting, long-distance running, or even extreme sports [9]. These advancements show how

prosthetics have moved far beyond their original medical function and into tools of performance, empowerment, and innovation. [figure 1.4]



Figure 1.4: Ottobock sports prosthetics

1.1 Motivation

The motivation behind this thesis stems from the need to provide an affordable, user-friendly, and high-functioning bionic that can be used in developing countries where access to high-end prosthetic hands is limited by cost, availability, and maintenance requirements. To address these issues, this project focuses on creating a wrist-level below-elbow bionic hand that can be mass-produced at a lower cost while retaining advanced control features and mechanical functions. By

implementing hybrid motor systems, tendon actuation, embedded control electronics, and integrated sEMG sensing, the design aims to bring advanced prosthetic technology within reach of more users.

Integrating surface Electromyography (sEMG) signals into prosthetic hand control stems from the need to achieve a more precise, real-time, and non-invasive interface for decoding user intent. EMG captures the bioelectrical signals generated by muscle fibers during contraction, providing a data source that directly correlates with intended motor actions. Unlike conventional mechanical based control schemes, EMG-based control enables proportional and dynamic modulation of prosthetic movements, essential for replicating natural hand dexterity.

The user experience and comfort of prosthetic devices remains a key concern, and many users end up abandoning prosthetic hands because they are uncomfortable, cumbersome, or difficult to use in daily life. Another very common problem faced with high- and low-end prosthetic is its durability, with prosthetics breaking within weeks of purchase. The hand developed in this project is designed with ergonomic considerations, using lightweight materials, compact motor placement, and a socket-and-cloth interface that securely attaches to the residual limb without causing discomfort. The inclusion of passive return mechanisms using rubber bands behind each finger adds to the smoothness of finger movements and another piece of rubber is also added between each joint to simulate the damping effects of human cartilage, reducing impact stress. Each component inside the palm has its own housing for protection and an additional cover was placed in the front and back of the palm to offer extra protection to the components inside. A glove-like cover was also placed on the entirety of the hand with damping material to cushion blows and reduce the chance of any part of the breaking.

From a research and engineering perspective, this project also presents an opportunity to explore multiple engineering approaches, combining mechanical engineering, biomedical signal processing, embedded systems, and control theory. It seeks to contribute not just a functional prototype, but also a design framework and methodology that can be adapted or built upon for future prosthetic technologies and low-cost industrial manufacturing.

Advancements in biomedical engineering, additive manufacturing, low-cost electronics, and embedded systems have opened new opportunities for creating more functional and cost accessible bionics which gives a chance for individuals to work on development without company funding. The development of tendon-driven mechanisms has been particularly promising, as it allows a more natural replication of the anatomical structures and motion pathways of the human hand. By mimicking the way muscles pull tendons to initiate movement, tendon-driven prosthetics can achieve smoother and more precise actuation compared to rigid mechanical systems such as linkage driven mechanism.

1.2 Aim of the project

The primary objective of this thesis is to design, develop, and validate a low-cost advanced tendon-driven bionic hand capable of independent finger movement. The proposed device should replicate human-like finger motions using tendon actuation and employ a hybrid actuation system that combines servo motors and geared DC motors. Actuation should be controlled using sEMG sensors with a threshold-based processing algorithm optimized for real-time embedded systems, eliminating the need for complex machine learning models.

This configuration will allow the user to control their prosthetic fingers using the brain computer interface set up in the bionic, where the EMG signals captured would decode the user's intent and intuitively allow multiple grip patterns.

For the index, middle, ring, and little fingers, a geared DC motor coupled with a pulley will be used to pull wire enabling individual control of each finger. As for the thumb, the combination of the servo and the geared DC motor would allow two degrees of freedom, replicating more natural and adaptive hand functions.

To simulate cartilage and absorb impact, rubber pieces will be incorporated between the joints, while low creep elastic rubber bands placed behind each finger will serve as a return mechanism. This would allow the fingers to return to their original position after flexion and allow the grip to be released in case of a malfunction that turn off the motors.

The hand will also feature a high-performance compact microprocessor that integrates all necessary signal processing, and motor driving electronics within the hand itself. Surface EMG sensors will be embedded into a forearm fabric sleeve for comfort and the sensor itself will allow real-time control of finger motion using muscle signals, while advanced filtering will be applied to eliminate noise and provide clean, ready-to-interpret data for precise and timely finger control. The final design must be self-contained, user-wearable, and attachable to the human forearm through a socket and fabric-based suspension system, with added structural protection and system-wide damping to enhance durability and reliability.

In addition to building the prototype, this project aims to establish a practical design methodology that other researchers and developers can follow in the creation of similar devices. This will include comprehensive documentation covering the mechanical assembly, control logic, electrical schematics, and data acquisition processes involved in EMG signal interpretation. Prototype validation will be carried out through functional testing to assess movement precision, response time, and overall comfort during use. By addressing both engineering performance and user-centered design principles, this work seeks to bridge the gap between academic innovation and practical, real-world prosthetic solutions.

Looking forward, potential future improvements may include the integration of pressure sensors on the fingertips to provide feedback for finer control over motor actuation and individual finger torque, particularly when holding objects of varying sizes, textures and fragility. When combined with the control input from the sEMG sensors, this additional feedback could enable more natural and human-like interaction with the surrounding environment. Furthermore, with access to medical laboratories and surgical expertise, the system could eventually be upgraded to use intramuscular electrodes, which offer better signal fidelity than surface EMG. In cases of recent amputations, nerve rerouting surgical procedures could be used to connect the remaining nerves that originally led to the hand directly to external sensors on the prosthetic surface, potentially restoring a degree of sensation to the user.

Chapter 2

Literature Review

2.1 Prosthetic Hand Development, Background and Problems

The field of prosthetic hand development has undergone significant evolution over the past century. From the earliest designs, consisting of cosmetic replicas, to the emergence of myoelectric devices, the trajectory has been marked by attempts to restore form and function to amputees. Early prosthetics, such as those used in World War I and II, were often simple tools made of wood or metal and operated by body-powered harnesses and provided basic grasping capabilities but were limited by their mechanical simplicity and the physical strain required to operate them.

The late 20th century witnessed a paradigm shift with the introduction of electromyography (EMG) signal control also known as myoelectric control, where electrical signals generated by residual muscles are detected by sensors placed on the skin and decoded to translate the user's intended movement. Devices such as the Ottobock Michelangelo Hand and the Bebionic hand represent this advancement, offering multiple grip patterns and adaptive control features. Despite their sophistication, such devices remain prohibitively expensive for many users and often require extensive training to master [10].

Parallel to commercial developments, academic research has focused on creating more anatomically and functionally accurate prosthetic hands. Projects like the LUKE Arm and the DEKA Arm have explored multi-degree-of-freedom (DOF) control, haptic feedback, and modular

architectures [11]. More recently, 3D printing has enabled the public use of prosthetic designs, allowing customizable and low-cost hands such as the e-NABLE and Open Bionics Hero Arm [12]. These designs emphasize accessibility but often compromise durability, fine motor control and automated finger control. The Hero arm, for example, uses EMG sensors to detect movement but requires pressing a button to alternate between different grip patterns. Other hands such as Psionic Ability Hand can provide these same features using a mobile application [figure 2.1].

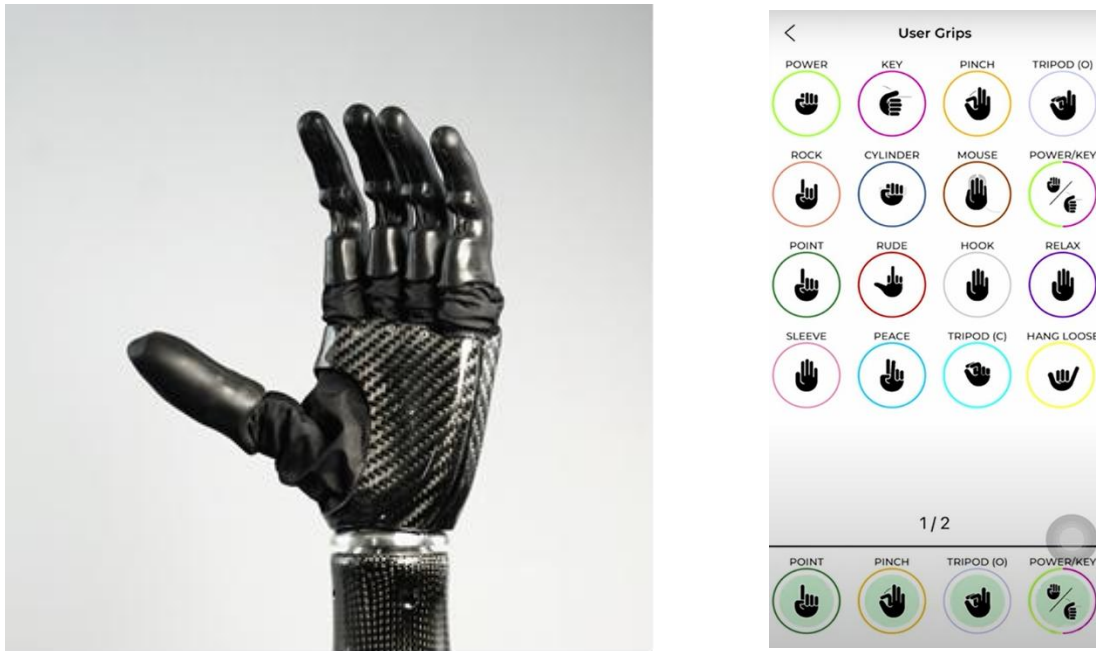


Figure 2.1: Psionic Ability Hand and mobile application

Advanced commercial prosthetic hands, such as the Össurs' i-Limb Quantum and Ottobock's BeBionic, offer a variety of grip patterns such as pinch, tripod, key, and power grips. However, similar to the bionic mentioned above, many of these grips can only be selected using a mobile application or button input, rather than dynamically responding to user intent in real-time through EMG signals which restrict spontaneity.

Despite the efforts of hundreds of companies and individuals, Challenges still remain in achieving lightweight, compact designs that support independent control of each finger. Simple research on most bionics sold showed that the most common problem faced lies in the three problems mentioned above. Cost, durability and automated independent finger control. Another major problem is prosthetic's ability to adapt to different users.

The last problem mentioned above is critical to consider as not all humans produce the same electrical signal strength. Factors such as electrode placement, skin sweat, muscle degradation are all factors that significantly affect the readings and the bionic might not work as intended, if proper control techniques are not used. Research has been made exploring the different techniques such as reinforcement learning and direct control showing that direct control provided better results [13]. Nonetheless, there is still a gap between laboratory prototypes and user-friendly, commercial-grade prosthetics that support both intuitive control and multiple degrees of freedom.

2.2 Prosthetic hand actuation mechanisms

Before discussion EMG signal acquisition and control, it is essential to understand the mechanism behind moving bionic hands. The two most commonly used types being linkage, tendon driven mechanisms.

Linkage driven mechanisms use kinematic analysis of linkages to create a mechanism that can replicate the natural movement of the human hand. In essence the design of any mechanism consists of rigid bodies called links connected to each other using joints. Using complex calculations, the links can be adjusted to enact a specific motion when provided with an input force whether it is rotary or linear.

This type of actuation provides high precision, repeatability, and ability to achieve controlled motion without requiring complex electronics or feedback systems. Unfortunately, this type of actuation can be expensive as the manufacturing of small links and joints requires high end CNC machinery. It can also lack in providing smooth human like movement and will require constant oiling to reduce friction between joints and prevent wear which will quickly lead to mechanical failure.

The ILDA hand is an example of linkage driven dexterous anthropomorphic robotic hand, achieving 15-DOF movements with 20 joints using four bar mechanisms, prismatic-spherical-universal (PSU) and prismatic-spherical-spherical (PSS) chains, bell cranks and linear motors. The hand was developed for robotics, but similar designs are often used in bionics [14] [figure 2.2]. A similar mechanism was developed during this research which will be mentioned in the experimental section below.

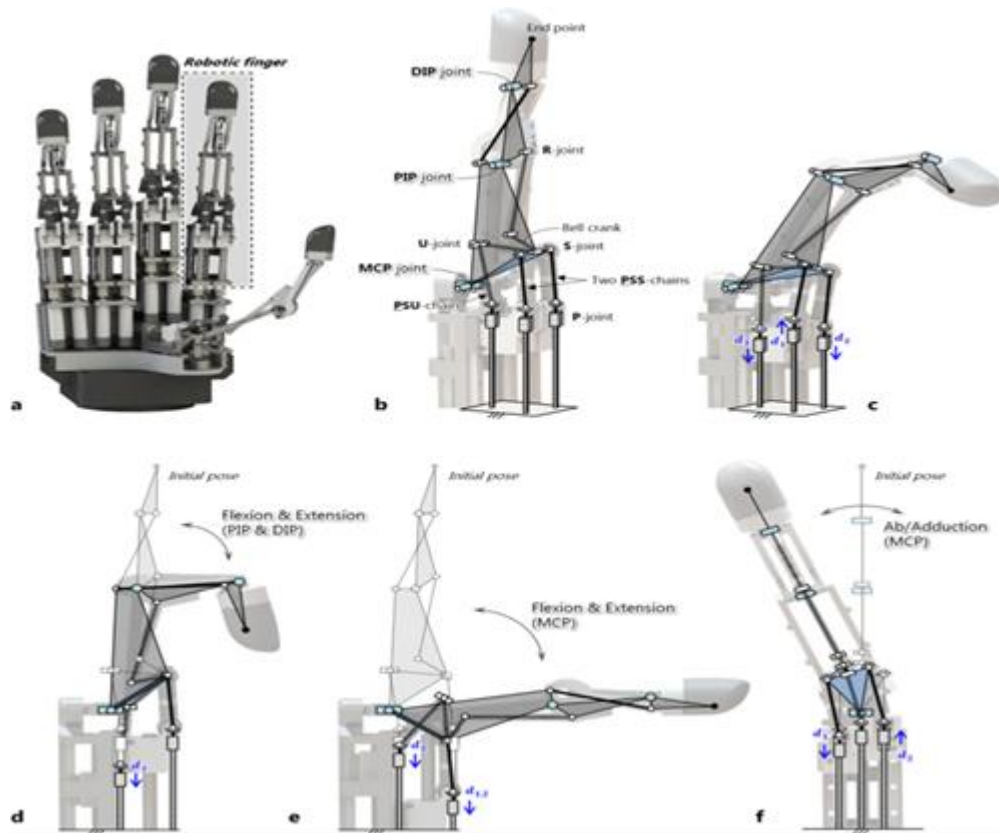


Figure 2.2: ILDA kinematic structure

Another example is the open-source anthropomorphic HRI hand. Providing an under actuated mechanism using two four bar mechanisms providing 15 degrees of freedom with 6 linear motors. In the context of robotic or prosthetic hands, an underactuated hand means not every joint is controlled by its own dedicated motor or actuator. Instead, some joints are mechanically coupled and move in coordination due to shared actuation mechanisms. This saves space and enables adaptive grasping but faces limitations when it comes to operations requiring precision. [15]

Tendon driven mechanisms, on the other hand, use a more biologically inspired approach to prosthetics. The use of metal or synthetic polymer strings connected to motors to move bionic fingers mimics the way the human hand uses tendons connected to muscles in the forearm to move the fingers. Pulleys and guides are often used to route the strings from the motors to the finger, which allows motors to be placed far away from the palm, often placed in the wrist, giving more space to add additional motors and provide more degrees of freedom.

Underactuated tendon-driven hands are particularly common, as they reduce weight and cost while enabling adaptive grasping. Elastic elements such as springs or rubber bands are often added to complement tendons by providing a passive return mechanism and impact damping, mimicking the cartilages and ligaments found in biological joints. An example of under actuation in tendon driven bionic is the SoftHand pro that uses a single actuator and 19 independent joints and still manages to naturally conform to the shape of objects, improving grip stability without requiring complex sensing or control algorithms [16].

Tendon slack, friction, and creep are common problems found in tendon driven prosthetics can lead to imprecise control and unpredictable behavior. For these reasons, the string material needs to be chosen carefully, and appropriate tension needs to be maintained. Modelling the exact behavior of tendon driven systems can also be an obstacle due to lack of engineering software and complexity equations, with several articles focusing solely on presenting systemic dynamic equations for modelling [17].

Very recently, a light weight, highly dexterous bionic was developed and represented a significant advancement in prosthetic technology, aiming to closely replicate the dexterity and functionality of the human hand, integrating 38 shape-memory alloy (SMA) actuators to achieve 19 degrees of freedom (DOF), enabling precise control over five fingers and the wrist. The device can perform 33 standard grasping modes and 6 advanced ones, enabling users to engage in a wide range of activities of daily living, such as writing and hair combing [18] [figure 2.3].



Figure 2.3: Prosthetic hand dexterity

Some designs combined both tendon and linkage driven mechanisms together to create a hybrid mechanism that leverages the compliance and adaptability of tendons alongside the precision and structural stability of mechanical linkages. The MCR-Hand II successfully performed multiple grasping poses including power grasps and precision grasps, imitating all DOFs (Degrees of

freedom) and function of a human hand [19]. Unfortunately, the hand is developed as an anthropomorphic robotic hand not as a bionic.

2.3 Prosthetic Hand Actuators

Apart from actuation mechanisms, actuators need to be taken into consideration. Both linear and rotary are viable options depending on the space available. With rotary actuation, RPM and torque need to be studied carefully to provide the hand with enough speed and gripping force. Similarly, stroke and force are accounted for in linear actuation.

One of the most commonly used actuators is the Geared DC Motor which can be used as both rotary and linear actuator to drive gears or pulleys as well as lead screws. Gears are used to lower the motor's speed and increase torque depending on the need. A problem is often faced because the motor requires an H-Bridge to control its speed and an encoder to determine the motor's position. This is a problem mainly faced in designs with compact spaces.

Servo motors are the same as geared DC motors, but they provide built-in position and speed control. The servo's rotation is often limited to 180 degrees, but servos with continuous rotation can also be found. While they can be found in several sizes and are widely available, micro servos have very little variety in torque compared to geared DC motor

Servo motors, especially micro servos are the same as DC motors but provides built in position control and are often used in fingers requiring angle-specific motion, such as the thumb. The servo's rotation is often limited to 180 degrees, but continuous rotation can also be found. While micro servos are widely available in a variety of small sizes, they have very limited torque compared to the geared DC motors.

Linear actuators generally work by the same mechanism mentioned above using DC motors and a lead screw, but they sometimes operate using hydraulic pump and pneumatic pumps. Unfortunately, they are very expensive compared to geared and servo motors and they aren't as readily available.

More recently, Shape Metal Alloys (SMA) are being used as linear actuators to drive prosthetics without taking much space. These alloys are made from materials that can undergo deformation (bending, twisting) and return to their original shape when exposed to the right temperature. With

a small amount of current they can be heated up to pull a mechanism and achieve high power while still remaining very light, achieving a pulling force of 70N [18].

2.4 Biomechanics

Studying the human hand and its biomechanics inspired the design of bionics. Closely analyzing the structure and function of the bones, muscles, tendons and cartilages of the hand helps in creating more anthropomorphic prosthetics that mimic nature's design.

Each finger is made out of three phalanges (Bones) except the thumb consisting of two. Between each of these phalanges is a joint. The top joint of the finger is called the distal interphalangeal (DIP), the middle joint is the proximal interphalangeal (PIP) and the joint between the finger and palm is the metacarpophalangeal (MCP). As the thumb only has two phalanges the topmost joint is called interphalangeal (IP). All the interphalangeal joints (IP, PIP & DIP) are called synovial joints which work the same way as an engineering revolute/pin joint which allows rotation around a single axis. Metacarpophalangeal joints are called condyloid synovial joints are equivalent to a universal joint allowing rotation in two perpendicular axes. The cartilages at the joints reduces the effect of impact stresses which can be simulated in a prosthetic using rubber,

Motion in the hand is driven primarily by skeletal muscles located both in the forearm and within the hand itself. The extrinsic muscles, such as the flexor digitorum superficialis, flexor digitorum profundus, and extensor digitorum, originate in the forearm and attach through long tendons into the fingers, enabling powerful movements. In contrast, intrinsic muscles like the lumbricals and interossei facilitate fine motor control and precision grip.

Force transmission from muscle contraction to joint movement is mediated by tendons, which act as elastic elements that transmit the pulling force to the bones, initiating joint movement. This inspired the design of tendon driven prosthetics that use strings operated by motors to pull fingers.

The tenodesis effect, a passive mechanism, is particularly relevant for prosthetic design as it describes how wrist extension causes passive finger flexion due to tendon tension, which can be exploited to design more natural prosthetic motion where pulling one string can cause several joints to move. The tenodesis effect is very similar to the concept of under actuation.

By replicating the biomechanical principles of the human hand, prosthetic devices can achieve both functional effectiveness and intuitive control. This biomechanical foundation is essential for

designing tendon-driven hands that offer both dexterity and adaptive grip patterns, closely mirroring those of the biological hand.

2.5 sEMG Signal Acquisition

Basic information about the physiology of the human nervous is required to know how to operate EMG sensors. The human muscles contract by a flow of ions in specific areas causing a difference in charge called action potential. When relaxed the muscles have a resting potential of -85 to -90 mV and These values can fluctuate in a process called graded potential where the voltage can increase up to -55 mV without causing a muscles contraction. Once the voltage reaches what is called the threshold potential (-55mV) the action potential starts where depolarization occurs causing a spike in voltage with its maximum value being around +30 to +50 mV. Then the value goes down in a process called repolarization where it drops below the resting potential and then back to its normal resting potential values. This whole process takes around 10 mS and the muscles contracts directly after the repolarization [20].

The strength and speed of all impulses are almost constant with very slight variations. The frequency of the of impulses in what varies, causing contractions of different strengths. For example, the grip required to hold a pen or lift a dumbbell will have the same maximum voltage value of action potential. What will change will be the frequency of impulses sent to the muscle [figure 2.4].

Wave summation

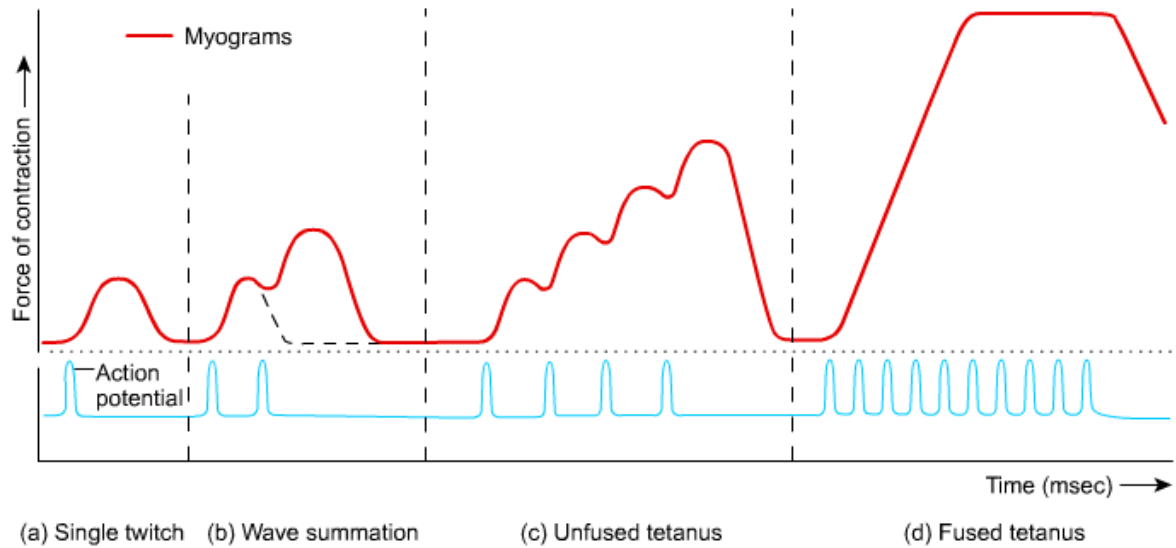


Figure 2.4: Graph of action potential and muscle contraction

Surface electromyography (sEMG) is a non-invasive technique used to measure the electrical activity mentioned above. It provides a reading of the small changes that occur and with some filtering can produce a signal that can be processed and with some control techniques or machine learning, it can be used to control different motors, enabling intuitive control.

sEMG signals are typically collected through electrodes made out of Ag/AgCl gel based or dry, placed on the surface of the skin above the targeted muscles groups. The signal then passes through different filters to remove noise. Notch filters are used to remove electrical noise, Bandpass of 10-500 Hz bandwidth to remove low frequency movement artifacts and high frequency noise, Highpass to remove the effect of the movement of electrodes. Rectification is then used to convert all negative signals to positive and an amplifier with 60 dB gain to increase signal strength as it

typically ranges from 0 to 5mV. With all that taken into consideration factors like electrode configuration and muscle anatomy can still influence signal quality [21][22].

Several factors can still affect the sEMG signal such as ambient noise from electromagnetic radiation which can be one to three times higher than EMG signal, motion artifact from electrode interface and cables and has a voltage comparable to the amplitude of the EMG, the instability of the signal obtained from the human muscles and the inherent noise in electrical components. [23]

The electrode placement strategy varies depending on the muscles being monitored. The flexor carpi radialis, extensor carpi radialis, and flexor digitorum superficialis are commonly targeted due to their role in wrist and finger movements, with the digitorum superficialis providing the highest EMG voltage reading. Medical professionals or trained technicians often assist in locating optimal electrode positions to ensure reproducible and reliable measurements across sessions and users [24] [figure 2.5].

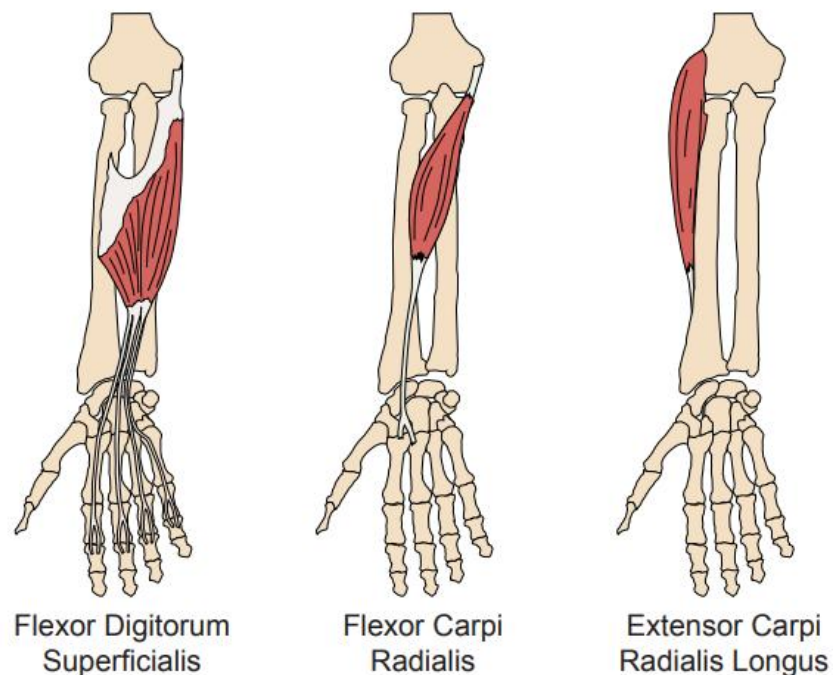


Figure 2.5: positions of targeted muscles in human hand

Basic sEMG electrodes usually consist of three electrodes, one electrode is used as a ground and should be positioned on a neutral bony area with no electric signals with the most common placement for this electrode being the elbow. The other two electrodes should be placed 2 cm apart to ensure high signal quality [25], one centered on the belly of the chosen muscle and the other aligned with the muscle fibers. Ground electrode is often yellow in color, and the other two are red and green.

2.6 Control Strategies

Control strategies for EMG signals involve processing the electric signals detected from the muscles to extract meaningful features and translate them into movements of fingers. The aim of this chapter is to review key approaches to EMG control and highlight their respective advantages and limitations.

On/Off control detects whether the EMG signal amplitude crosses a predefined threshold. Falling above or below that threshold switches between on and off state. This method requires very low computational power but can be subject to false positives if the threshold value is low. Some prosthetics were developed so the threshold value can be adjusted by the user, using an externally placed knob [26]

Proportional control extends on/off control by adding a control parameter to achieve more natural control. The higher the muscle contracts the greater the output would be thus allowing the user to control the grip force. The problem with proportional control is that it makes the user susceptible to muscle fatigue, which can alter signal amplitude and lead to inconsistent control [27].

Pattern recognition strategies represent a significant advancement, allowing for the classification of distinct muscle contraction patterns by interpreting connections between different muscle signals and then predicting user intent. This is done using multi-channel EMG signals including time-domain, frequency-domain, or time-frequency features. These features are then fed into a machine learning algorithm such as Discriminant Analysis (LDA), Support Vector Machines (SVM), Artificial Neural Networks (ANN), k-Nearest Neighbors (k-NN), or more recently, deep learning models, all trained to recognize specific movement intentions such as movement of index finger alone [28]. This method still faces a lot of problems, mainly, the extensive computational power required, and the training data needed for the models.

Regression-based control is similar to pattern recognition, but it tries to predict kinematic or kinetic parameters such as joint angles and forces directly from EMG signals, which can provide a more natural control of prosthetic joints. This control method faces the same problems as pattern recognition and can also be more sensitive to noise and signal variation [29].

Hybrid control strategies combine elements from different approaches to leverage their strengths and mitigate their weaknesses. This often involves integrating pattern recognition for mode switching with proportional control for fine-tuning. Other hybrid strategies include a combination of EMG and electroencephalogram (EEG) [30]. Hybrid system can have the best potential for providing intuitive and flexible user interaction, but they require very complex systems and highly sophisticated signal processing.

Chapter 3

Methodology

Due to the absence of high-quality low-cost bionics in the market in Egypt, designing a bionic was imperative for proper testing. The prosthetic was designed to be compact with all components present in the palm. After the design and assembly of the bionic, EMG sensors were used for control of the hand. Simple methods of signal processing were not producing accurate readings, instead an adaptive control algorithm with dynamic thresholding was used to control the mechanism, which showed significant improvements.

3.1 Development of Linkage Driven Mechanism

Initially, focus was directed at linkage-based mechanisms as it can be studied to produce predictable movements and had less complications compared to tendon driven mechanisms especially regarding tendon driven mechanisms need for tensioning and small design tolerances.

Linkage Mechanism Design and Simulator was used to design a planar linkage mechanism with multiple links and connectors, designed to mimic the motion of a human finger, including the Distal Phalanx, Middle Phalanx, Proximal Phalanx and 2 linear actuators. Through calculations, link length ratios, and Grashof's Law, a hybrid structure combining four-bar linkages and slider-crank elements with prismatic joints was designed, enabling controlled flexion and gripping.

Average ratio of each phalange was taken into consideration to present a design that resembles a human finger. Especially the ratio between the middle and proximal phalange as these cannot be modified later. The distal phalanx was not considered as it would be modified to proper length during design. The average ratio of a human hand proximal phalanx to middle is 1.64:1 and the model was designed with ratio of 1.62:1 to mimic near-anatomical proportions. The length of each link was designed in accordance with Grashof's law, and the length of each link was modified to perform movement similar to human hand. The dimensions are mentioned below, and the design can be seen in figure 3.1.

Table 3.1

Link	Dimensions/mm
Distal Phalanx	30.00, 7.50, 30.89
Middle Phalanx	48.30, 41.39, 12.89
Proximal Phalanx	79.07, 9.60, 62.26, 36.60
Proximal Phalanx Coupler	17.36
Distal-to-Proximal Coupler	51.24
Middle Phalanx to Proximal Coupler	54.570

$$Q = 49.41 \text{ mm}$$

$$S + L = 17.36 + 79.07 = 96.43$$

$$P + Q = 62.49 + 49.41 = 111.90$$

Since $96.43 < 111.90$, the linkage satisfies Grashof's Law, classifying it as a crank-rocker.

The loop-closure equation is for the four-bar mechanism:

$$r_1 e^{i\theta_1} + r_2 e^{i\theta_2} = r_3 e^{i\theta_3} + r_4 e^{i\theta_4}$$

Equation 3.2

Where:

r : Length of Link

θ : Counter Clockwise angle between link and x-axis

$$r_1 = 79.07 \text{ mm.}$$

$$r_2 = 45.94 \text{ mm.}$$

$$r_3 = 17.36 \text{ mm.}$$

$$r_4 = 36.60 \text{ mm.}$$

Numerical methods such as the fsolve function in MATLAB can solve for joint angles given an input angle at MCP θ_1 for further study.

The finger was able to perform multiple different movements depending on input from the linear actuators. An equal input from both actuators performed a grip movement [Figure 3.2]. Input from first actuator and anchoring the second performed DIP and PIP flexion. [figure 3.3]. finally input in the ratio of 1:1.785 performs MCP flexion [Figure 3.4].

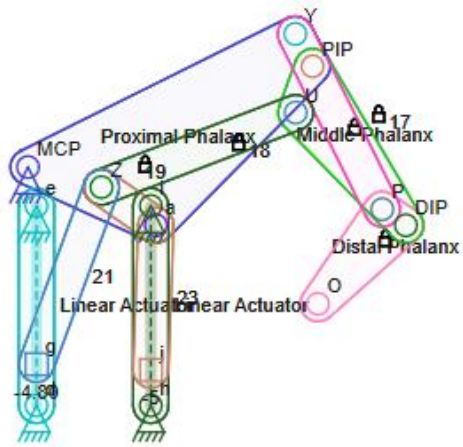


Figure 3.2: Grip

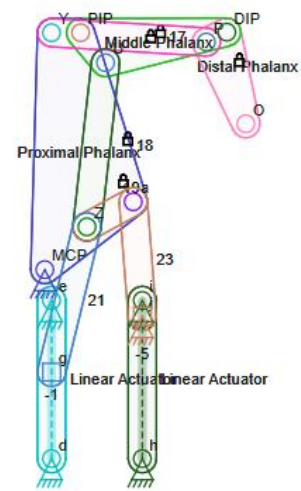


Figure 3.3: DIP & PIP flexion

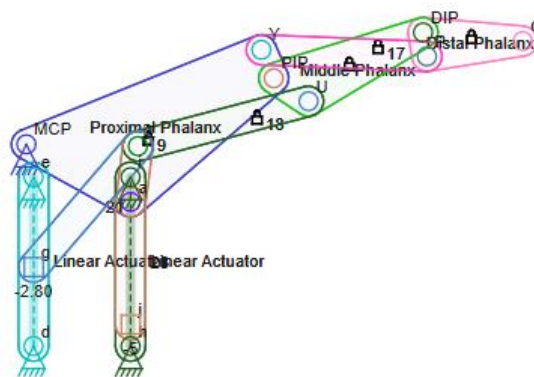


Figure 3.4: MCP Flexion

A Solid Works model was designed following the exact dimensions mentioned above and simulated using motion study, which showed the same movement simulated using Linkage Mechanism Design and Simulator. The design and simulation proved successful, but the prototype was abandoned due to difficulty manufacturing small enough metal links with good dimensional accuracy and enough strength to function similarly to the human hand. Another issue was the scarcity of linear actuators of proper compact size and force in the Egyptian market. This led to switching the design to a tendon driven mechanism.

3.2 Development of Tendon-Driven Mechanisms

The tendon-driven mechanism for the prosthetic hand was developed to replicate natural finger flexion and extension using a lightweight and compact actuation system. The process began with defining the desired range of motion, based on anatomical references for tendon placement, and each finger was designed to have independent tendon paths that allow controlled flexion through motor-driven actuation, and passive extension through elastic bands acting as return springs.

Fishing braid line from hegemon with eight strands of tightly woven fibers with a total wire diameter of 0.6mm was chosen as it provides a carrying capacity of 50Kg which is more than enough for each finger to carry object without snapping while still having a very small size.

Multiple prototypes were implemented in Solid Works to properly house each string pathway with each turn curved to prevent the wire from eating away at the housing with constant use. The string was connected to N20 motors via pulley placed on the motor shaft, then moves along the finger through smooth 1mm channel guide, around a ball bearing at the tip of the finger and back along the finger where it is fixed [Figure 3.5]. The ball bearing is 2mm in diameter and it was used to smooth movement of the string in the curved area. This mechanism was designed to half the torque on the motor, when force is applied to the wire by halving the force on the finger with half the tension of the wire going to the motor and the other half to a fixation in the bionic palm.

F: External Force

T1: Tension on wire connected to motor

T2: Tension in wire tied to a fixation

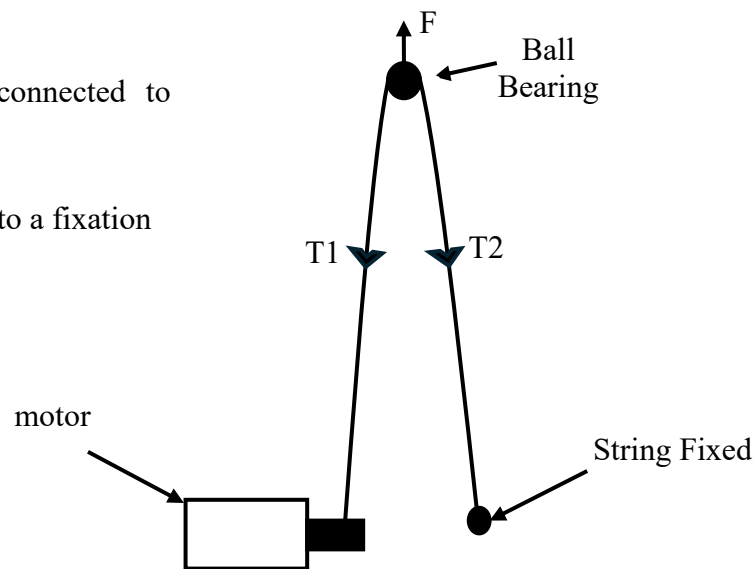


Figure 3.5: Mechanism to half torque on the motor

Elastic return bands were integrated at the finger joints to passively restore the fingers to the extended position when tendon tension was released. The bands were tested applying 10Kg of force for 1 week to test for creep effect and ensure the return mechanism will remain functional with constant use. The tension and attachment points for these bands were experimentally adjusted to optimize force of return and prevent overstressing the tendons and motor during flexion.

Prototypes were fabricated using 3D additive printing using PLA, and assembly was carried out manually. Several iterations were conducted to refine tendon path smoothness, reduce friction, and ensure durability of the guiding pulleys and anchor points

One of the major problems faced when developing a tendon actuation hand was finding an engineering software to test out the finger flexion and extension movements and ensure it is smooth. As a consequence, this too had to be carried out manually on a prototype.

3.3 Motor, Driver, and Battery Selection for Prosthetic Actuation

Choosing the right motor for the prosthetic was based on finding optimum force and RPM to mimic real life movement. The average human finger can flex in a time ranging from 0.1 to 0.5 seconds and assuming each finger can carry 3Kg, and calculations were used to find range for motor RPM and torque that can imitate these values taking into consideration the availability.

$$RPM = \frac{30S}{t\Omega r}$$

Equation 3.3

Where:

S = distance travelled by wire to achieve full flexion

t = time

r = radius of pulley

$$T = F \times r$$

Equation 3.4

Where:

T = torque of the motor

F = maximum force finger can withstand

r = radius of the pulley

Using the above equations and increasing ranges to acceptable values to match motor availability, with a distance s of 70mm and radius r of 4mm the motor should have RPM ranging from 1670 RPM for flexion in 0.1s and 111RPM for 1.5s. For carrying 3 Kg per finger, 11.76 N.cm of torque is needed.

N20 motor 12GAN20-200 geared motor was chosen as it provides 100RPM, 3.924 N.cm moving torque under rated load and 9.81 N.cm holding torque under 6V. This provided 1.67s flexion with 8Kg carrying capacity per finger.

The motor can be used as is to provide very high carrying force but very low speed or it can be modified by disassembling the gear box and removing one of the metal gears and adjusting the position of another to double the RPM and half the torque. Speed was chosen as the priority, and the modification was done providing 0.835s flexion with 4Kg carrying capacity which is within the range given above. With the mechanism mentioned above and shown in *figure 3.5*, the motor torque is doubled, giving 8Kg carrying capacity with the holding torque.

Motor drivers were chosen based on size and efficiency. TB6612FNG dual H-bridge motor driver IC was used for controlling DC motors due to its very small size and high efficiency compared to commonly used drivers such as L298N.

MG90s with metal gears were chosen for thumb movement due to its compact size and 2.5 Kg.cm holding torque which can rotate the thumb while maintaining a high enough torque to carry objects.

The STM32F401 “Black Pill” development board was chosen as the central processing unit for the prosthetic hand due to its high performance, size, and cost-effectiveness and very high processing power. Running at 84MHz, this microcontroller can handle real-time signal acquisition, processing, and motor control required for the tendon-driven prosthetic system with minimal delay.

To provide enough voltage to run all motors, servo and Black Pill, 2S Lipo battery was assembled using 2 Lipo cells connected in series to provide 7.4V and 1000 mAh capacity and the resulting battery was connected to Battery Management System (BMS), TP5100 module, PD Decoy Trigger Motherboard Module Type C USB, IRF540N mosfet and LM340T5 IC [Figure 3.6]. The first three components were used to allow battery to be charged safely using any type C charger and provide overcharge protection, over-discharge protection, short circuit protection. The NPN Mosfet protects the Black Pill and stops all hand movement when charging by cutting power to the Black Pill. This is done using chrg pinout from TP5100 module that provides either 0v or floating output.

The floating output was connected to a pull up resistor to allow it to be high and be used as a gate for the mosfet. Finally, the LM340T5 IC was used as a voltage regulator to provide the microcontroller with 5v input.

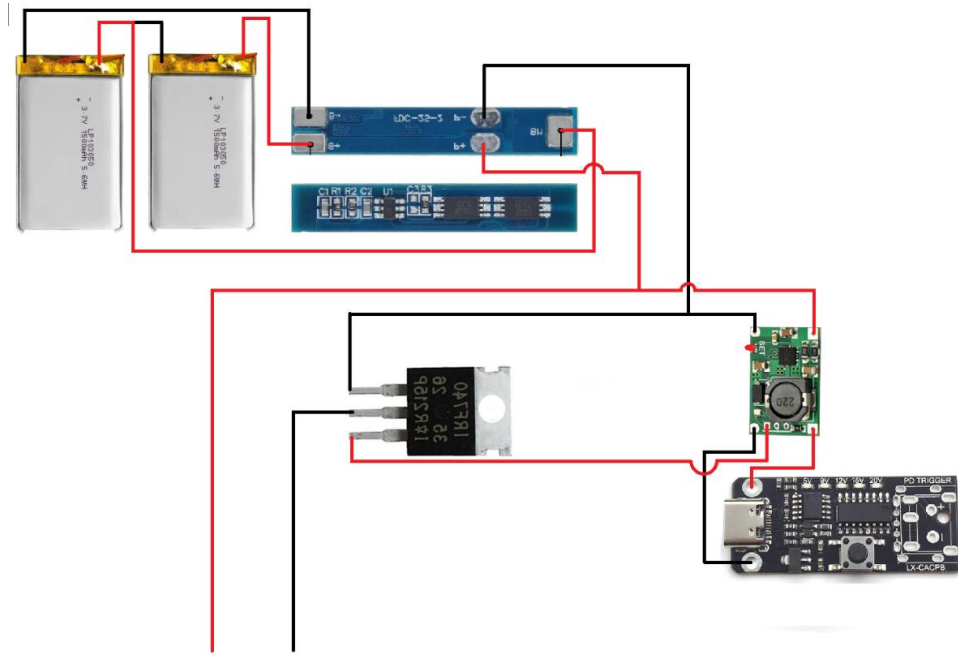


Figure 3.6: Battery connections

Ideally a battery with 3000mAh should be used to provide the hand with more runtime before needing to be charged but no such batteries were found in Egypt with a small size and enough capacity.

3.4 3D Design of the Prosthetic Hand

The design of the prosthetic hand involved the creation of eleven prototypes on Solid Works. The final design consisted of one hundred and fifty one components modeling all mechanical, structural, and functional components of the hand.

The design process started by defining the dimensions of each finger joint for an average adult male, in order to create an anthropomorphic model. Each finger joint was designed individually in reference to these dimensions. Each phalange was linked to the next using a pin joint and a rubber part was placed between any two parts that will touch to act as a damper shown in black in *figure 3.7*. The finger was designed to allow a maximum frontal rotation of 95 degrees and backwards rotation of 5 degrees.

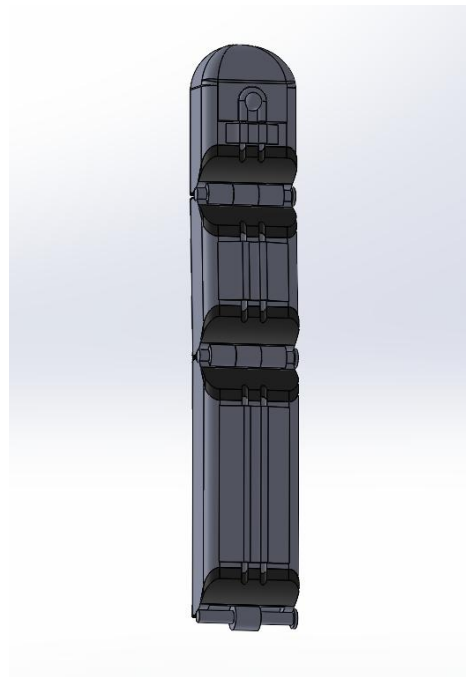


Figure 3.7: Index finger assembly

Along the finger can be seen two parallel guides, 1 mm in internal diameter, that connect at the tip of the finger. These guides are for the tendons to be inserted and move along the finger. At the tip

of the finger the tendon will rotate around a ball bearing 2 mm in diameter and go back down. The tip is made of two parts, the distal phalanx with the tendon guides and the ball bearing cover to be placed after the bearing is inserted [figure 3.8].

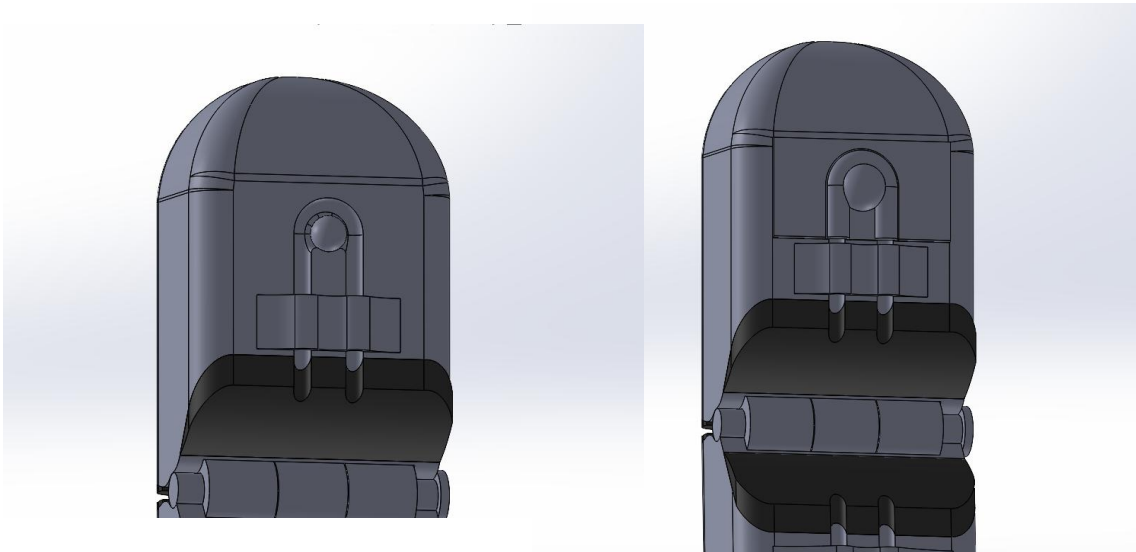


Figure 3.8: showing tip of the finger

The thumb is designed to house a servo that rotates it around an axis. The thumb and the housing containing the servo are both rotated by an N20 motor around the perpendicular axis and limited in rotation by a stopper [figure 3.9]. The housing contains small 1 mm holes to allow a tendon to pass through and the pin and joint mechanism the housing rotates around is connected to the hand using 3mm screws.

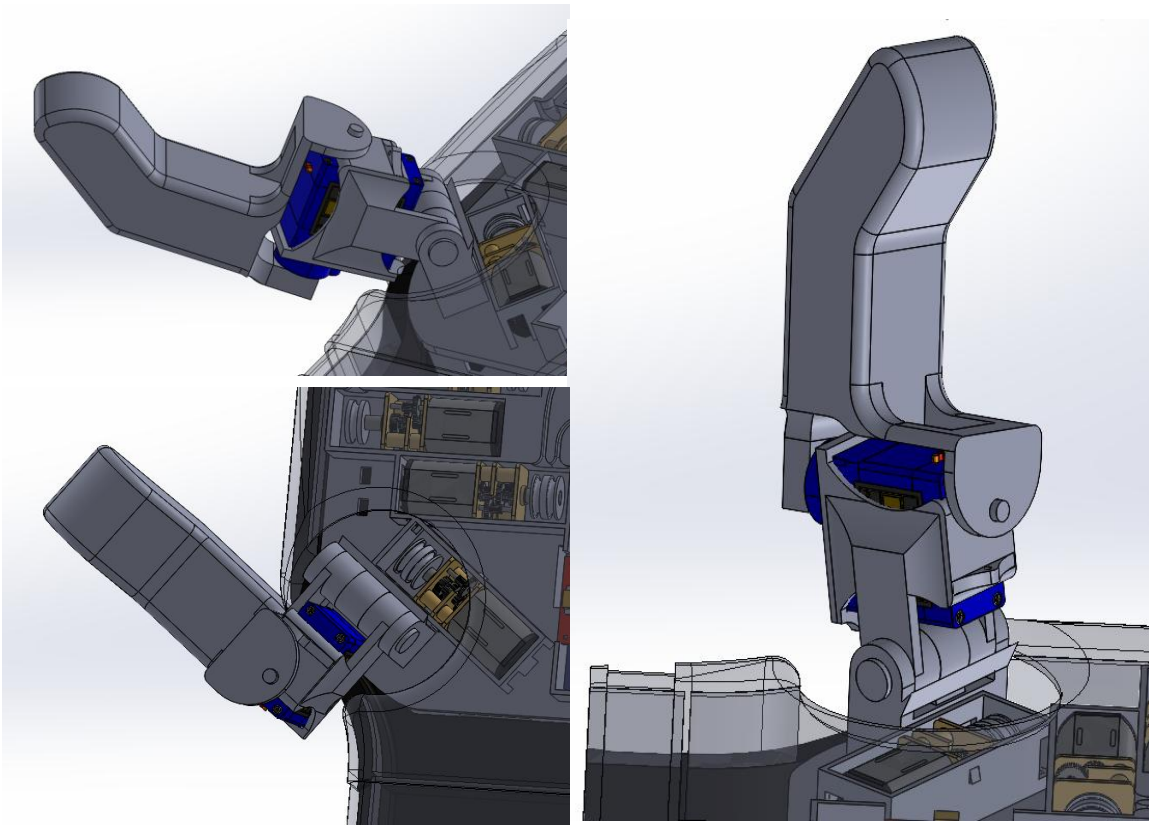


Figure 3.9: showing servo in different positions

The palm of the hand is made to house all components with a tight fit to prevent the motors from vibrating and moving out of place [figure 3.10]. Holes were placed strategically along the palm to allow wires to move to intended places without taking space. Circular extrusions can be seen in the back of the palm; this is designed to hold the back cover in place. The front and back covers were made to provide protection for all components inside, and were made to be held together using brass inserts and 3mm screws.

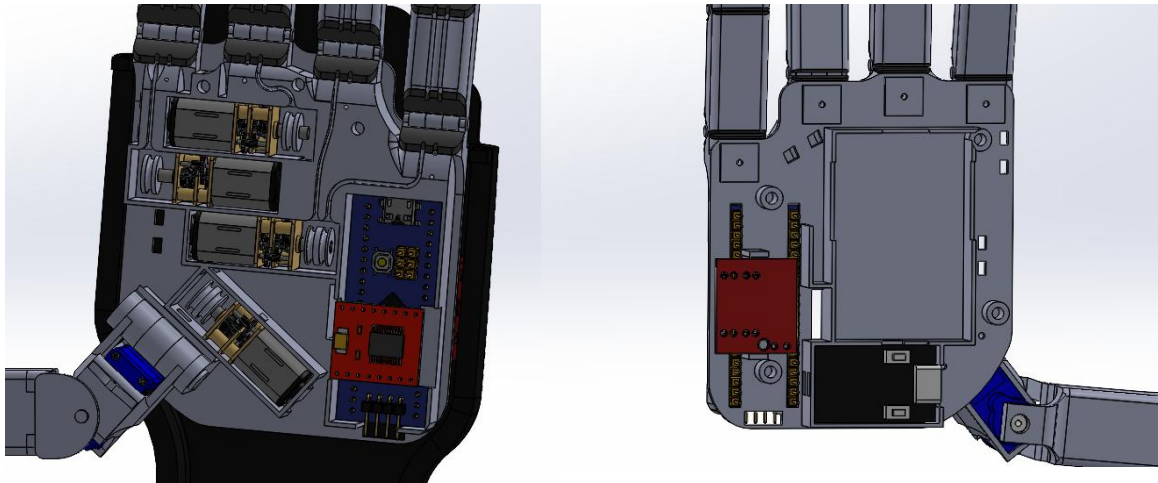


Figure 3.10: Front and back of the palm showing all housings

The hand was tested and was able to perform multiple grip patterns, imitating that of a human. The thumb was also tested to make sure it is opposable and can touch all four fingers [figure 3.11].

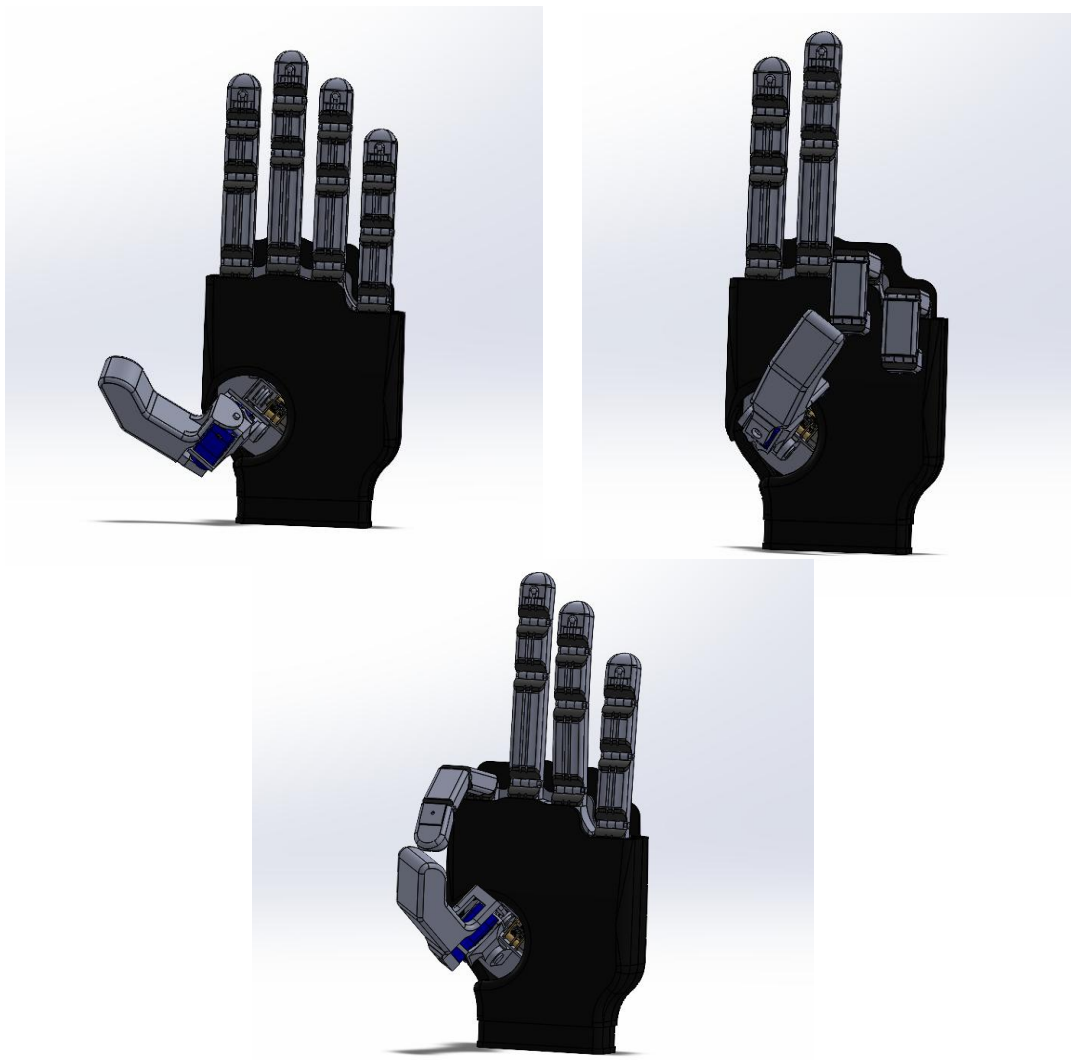


Figure 3.11: showing the hand performing multiple grip patterns

3.5 sEMG Signal Acquisition: Sensors, Muscles, and Noise Challenges

Selecting appropriate EMG sensors and understanding optimal sEMG electrode placement was crucial for acquiring clean signals to enable effective control implementation.

Two sEMG types were considered as options, wet and dry electrodes. Wet electrodes use a conductive gel placed beneath electrodes to improve signal quality significantly, but it can be very time consuming to apply, and the gel can dry out over time. Dry electrodes forgo the gel but provide more noise and motion artifact. Initially, dry electrodes were preferred for their ease of use, but due to limited availability, wet electrodes were selected.

Three Ag/AgCl electrodes (red, green, and yellow) were used to measure the voltage output of a selected muscle group. Red and green were placed 2 cm apart on the chosen muscle group for best signal reading and the yellow was placed on the elbow as it's a bony part with minimal electrical activity to serve as a ground. Before placing the electrodes, conductive gel was added directly on the skin to enhance conductivity and ensure accurate readings.

The Muscle Sensor V3.0 was connected to the electrodes via a 3.5mm audio-style male jack to filter the signal as it employs a multi-stage filtering process to convert raw EMG signals into a clean, usable analog output proportional to muscle activity. The signals captured via three electrodes range from 10 μ V to 2 mV, with a frequency of 10–500 Hz and is amplified by the module using an instrumentation amplifier with a high common-mode rejection ratio >90 dB to reject noise like 50/60 Hz electromagnetic interference. A high-pass filter cutoff 10–20 Hz removes low-frequency motion artifacts, while a low-pass filter with cutoff of 500 Hz attenuates high-frequency electromagnetic interference, together forming a band-pass filter that isolates the EMG signal's frequency range. The signal is then rectified using a precision rectifier circuit to convert the bipolar AC signal into a unipolar signal, followed by a smoothing low-pass filter with

a cutoff of 1–10 Hz that provides the rectified signal into a stable DC-like output. A final amplification stage adjusts the signal to a range of 0 to +3.5V, suitable for processing.

The Muscle Sensor V3.0, the only model available in Egypt, was used despite being discontinued due to the availability of newer models with improved functionality. Its primary limitation is that it requires 18V to operate a single sensor, in contrast with later models such as the MyoWare 2.0 Arduino Shield which can operate using 5V or 3.3V and can connect to six sensors. The high voltage need required the EMG sensor to be placed externally as it needs to be connected to two 9V batteries in series which produces nearly the output of the 2S battery used.

Initially, electrodes were to be integrated into a cotton, polyester and spandex fabric to reduce motion artifact as it is still an issue the muscle sensor cannot filter out. The elastic nature of the fabric would be tight enough to hold electrodes in place during movement while still being comfortable enough to wear on a daily basis. Unfortunately, this couldn't be implemented with the given timeline as the development and fabrication of such a prototype would take a long time to develop and test out.

The sEMG was to be tested on six different muscle groups simultaneously. Flexor digitorum superficialis lateral to profundus in mid forearm provided the highest voltage and it indicates flexion of index, middle, ring and little fingers, and the flexor digitorum profundus from the medial side provides the same reading with lower voltage. Extensor Digitorum Communis gives reading related to extension of index, middle, ring and little fingers and Extensor Digiti Minimi provide data about extension of little finger. Finally, Flexor pollicis longus and extensor pollicis longus provide signals for flexion and extension of thumb respectively [figure 3.12]. With advanced signal processing and combined targeted electrode placement it is possible to isolate finger-specific muscle activity by mapping each finger to specific combination of signal from all six sensors.

Muscles of the Forearm (right arm)

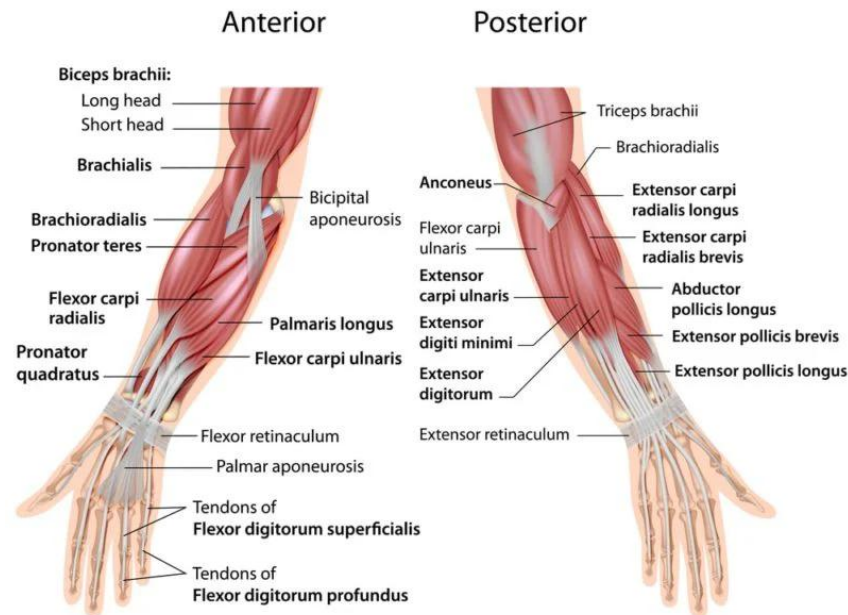


Figure 3.12: Muscles of the forearm

Implementing six sensors for the set up mentioned above was not possible due to the lack of modern EMG sensors, and the prohibitively expensive price of the muscle sensor V3.0.

3.6 Adaptive Control Algorithm for EMG-Based Actuation

After acquiring filtered and clean readings from the Muscle Sensor V3.0, real-time adaptive control algorithms run on the microcontroller to process the inputs and adapt to variations to ensure noise tolerance across different muscle groups and muscle users, providing a binary actuation of either on or off.

The algorithm starts with a five second calibration phase during which the user is instructed to keep their muscle relaxed and an LED blinks to alert the user of the calibration stage. During calibration, a buffer is used to hold five hundred samples, which are collected at 100 Hz using an Analogue-to-Digital Converter (ADC) pins on the microcontroller and are then converted to voltage values by using a linear scaling factor where 5 V corresponds to the maximum ADC reading of 4094. The voltage is then checked to be within a range bigger than 0.1V and smaller than 80% of the maximum voltage in order to avoid any extremes. Voltage is then smoothed using Exponential Moving Average (EMA), with an EMA factor of 0.2 and placed inside the buffer. EMA is similar to a normal mean, but it adds weight to older value in order to combat the fluctuations and while still making it responsive to change.

$$V_{smoothed(n)} = \alpha \times V_{current} + (1 - \alpha) \times V_{smoothed(n-1)}$$

Equation 3.5

Where:

$V_{smoothed(n)}$: New value of smoothed voltage

α : EMA factor

$V_{current}$: Current voltage value captured by ADC

$V_{smoothed(n-1)}$: Previous value of smoothed voltage

After the smoothed voltage is calculated, it is inserted in a buffer and sorted using a bubble sort algorithm in order to find its median. This median serves as an initial baseline voltage which represents the resting EMG signal output of the user. As a fail safe for the calibration in case the user flexes his hand or the electrodes were placed incorrectly, the maximum and minimum values of the buffer are stored and if their difference is bigger than 0.2V, the calibration starts again.

After calibration is complete, variance around the baseline is calculated to measure how far the value spread around the baseline voltage. This is done using the same buffer used in the calibration process. The variance is expected to be low but in the case of a very stable output, if the variance is less than 0.001, it will be to 0.001.

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (V_i - V_{baseline})^2$$

Equation 3.6

Where:

σ : Variance

N: Number of samples

V_i : Voltage value inside buffer

$V_{Baseline}$: Baseline voltage

Using the baseline, variance and constant K, the threshold of the signals is calculated. The threshold voltage is the value that differentiates between the relaxed and contracted state of the muscle. Constant K is a multiplier that scales the variance of the signal to set the activation threshold at a level that reliably distinguishes between muscle states and noise. After extensive testing and fine-tuning of different K values, 5 consistently delivered the highest accuracy and reliability. After calculation, the threshold is checked to ensure it remains smaller than or equal to 0.5 V above the baseline.

$$V_{threshold} = V_{Baseline} + K \times \sqrt{\sigma^2}$$

Equation 3.7

Where:

$V_{threshold}$: Threshold voltage

$V_{Baseline}$: Baseline Voltage

K: Threshold multiplier

σ : Variance

To ensure the stability of the output and its adaptability to changes, all calculations mentioned above are continuously done after the calibration is complete, using a fifty-sample buffer that updates in a first in first out manner, instead of the five hundred sample buffer used in calibration. This reduction was made to provide faster data processing and response to change for better real-time adaptability.

The final output of the algorithm is binary, meaning it has two distinct values, either “1” representing an on state or in this case muscle contraction, or a “0” representing an off state or muscle relaxation. This is done by hysteresis-based switching to ensure stable switching. Hysteresis based switching differs from normal switching as it uses a range above and below a certain threshold that needs to be reached in order to change the current state. This range is called a hysteresis band. Using normal switching can cause very rapid switching between states, while the hysteresis band creates a dead zone between the two states where no change can occur.

$$\begin{aligned} V_{on} &= V_{threshold} + H \times V_{threshold} \\ V_{off} &= V_{threshold} - H \times V_{threshold} \end{aligned}$$

Equation 3.8

Where:

V_{on} : Minimum voltage required to reach on state

V_{off} : Maximum voltage required to reach off state

$V_{threshold}$: Threshold Voltage

H: Hysteresis percentage

Many failsafe mechanisms were also applied to shield against unexpected variations and output stability. The buffer, for example, is made to only be updated if the smoothed signal is below the threshold to ensure long contraction doesn't shift baseline to a high value causing the system to be in a constant "On" state. Change in variance is also limited to $\pm 10\%$ of the previous variance in order to ignore rapidly fluctuating readings. Restrictions were added to state changes so that "On" state activation requires smoothed voltage to remain above V_{on} for 300ms and the same thing applies for the off state, this was later removed to provide a faster response. Remaining in the hysteresis band for five seconds will also cause the state to toggle in order to recalibrate baseline. Finally, a significant change in baseline above 0.2V will trigger recalibration to ensure stability.

This adaptive control algorithm uses EMA smoothing, dynamic variance tracking, and thresholding to detect muscle activity reliably. Hysteresis and debounce protect against noise and fail-safe logic ensures reliability. It requires no ML training, making it perfect for real-time embedded systems like prosthetic hands.

The algorithm created doesn't use normal control methods of that requires tuning gains and tracking continuous errors. The methods used do not use feedback but instead use statistical thresholds, making it simpler and more suited to binary actions and unpredictable EMG signals. It can be defined as a variation of a bang-bang (on/off) control mechanism with adaptive thresholding, where the activation threshold is dynamically computed based on the real-time statistics.

Chapter 4

Experimental work

This section outlines the practical implementation conducted to evaluate the functionality and performance of the proposed prosthetic hand system. The experimental work involved the fabrication assembly of all 3D printed components, integration of electronic systems, development of control algorithms, and real-time testing using surface electromyography (sEMG) signals.

The prosthetic hand was tested to ensure practicality for users, and the EMG control was tested for its accuracy and responsiveness to change. The results obtained from these experiments provide critical insights into the viability of the design for real-world applications and inform areas for further optimization.

4.1 Motor deconstruction, battery system connections and testing

Disassembling the N20 motors to reduce the gear ratio and increase speed presented a significant challenge due to the miniature scale of the gears. The process began by removing the screws securing the gearbox to the motor. The gearbox consisted of two press-fitted levels, which were carefully separated using a slotted screwdriver that was inserted at the seam and gently twisted to avoid damaging the gears or bending any components.

After detaching the lower section of the gearbox, the small gear meshed to the motor shaft pinion gear (seen in Figure 4.1) was removed. The gearbox floor was then reattached by pressing the parts back together using pliers. Before reattaching the gearbox onto the motor, a 3mm slotted screwdriver was used to carefully slide the motor's gear upwards along the shaft (as seen in Figure 4.2). This repositioning allowed the gear to engage with the second gear in the gearbox rather than the one that was removed.

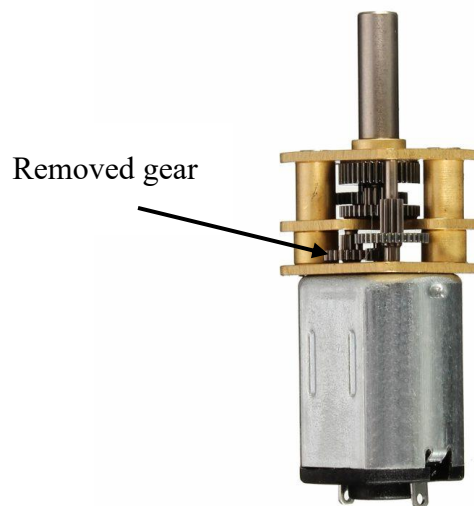


Figure 4.1: showing gear to be removed



Figure 4.2: sliding motor's gear upwards

Working with such small gears proved difficult, especially during the separation of the press-fit sections, as the tiny gears would often fall out and were difficult to relocate. Another major challenge was moving the motor gear upwards without applying excessive axial force on the shaft, which could damage the motor's internal stator. To prevent this, small pliers were used to hold the shaft securely in place while applying only the minimum force necessary to adjust the gear.

Following the completion of the motor gear ratio adjustments, the motors were connected with motor driver circuits for control. Two motors were connected to a TB6612FNG driver module. Due to the limited availability of only one TB6612FNG driver, the other two motors were connected to a TC1508A motor driver, which provided similar dual-channel H-bridge functionality. Both driver modules were then interfaced with black pill microcontroller via breadboard and tested with a standard 9 V battery to ensure motor functionality.

The battery connections, as shown in *Figure 3.6*, were soldered and securely wrapped in insulating tape to hold them in place. The battery was then used to power the motors, and it operated as expected. Subsequently, the battery charging functionality was tested using a Lenovo 140W USB-C laptop charger (20V, 3.25A), standard USB-C port from a laptop (5V, 0.5–0.9A typical) and a mobile Super-Fast Charger (Samsung 25W, 9V, 2.77A) and the PD Decoy Trigger Motherboard

module voltage regulation proved effective and charging proceeded as expected with a 9 V steady voltage to the batteries, measured using UNI-T UT33B+ digital multimeter, confirming system compatibility and stability. A short circuit test was also performed and the TP5100 module safety features engaged successfully, preventing damage to all components and verifying system built-in protection capabilities.

4.2 Using Real-Time EMG Signals

After confirming the functionality of all electronics, a simple code was used to identify average output the muscle sensor. The EMG electrodes were placed on the flexor digitorum superficialis as it produces the highest voltage output when flexing all fingers in a grip motion. Electrodes were connected to the Muscle Sensor V3.0, which was powered by two 9V batteries and connected to the Black Pill microcontroller.

As soon as the sensor was connected to the Black Pill, the microcontroller stopped working and with further testing it was discovered that the muscle sensor output was extremely high, reaching 9 V instead of the regular 5 V it should output. This was caused by a faulty amplifier, and it led to the Black Pill burning out.

The sensor was tested using the multimeter, and after extensive investigation and reviewing the sensor schematics, the problem was identified to be a faulty amplifier. Due to a limited timeline the Black Pill was replaced with an STM32F103C8T6 Blue Pill which was on hand and still operated the same way but with lower computational power. The connection between the signal pin and blue pill was protected using a voltage divider consisting of 330 Ω and 220 Ω resistors to reduce the 9V output to 3.6 V.

Even after the modification, the signal output was not correct. The voltage readings either oscillated between 0.002 V and 0.009, indicating no signal, or remained fixed at 1.25 V with no variation. Further testing of the sensor was conducted, using multimeter continuity test to ensure all connections were intact. One of the probes of the multimeter was placed at each end of the aux male connector, and the other probe was connected to the electrode end that attaches to the skin. The results were inconsistent and failed unless the electrode was pressed with high pressure.

To further investigate, the electrodes were placed on a piece of metal, and the test was repeated with one probe on the aux and the other at the far end of the metal piece and the test failed again. When the Ag/AgCl was removed the test was repeated with the sensor bare metal it passed successfully, indicating the problem was the Ag/AgCl connections.

New, higher-quality connectors were acquired, along with conductive gel, and the continuity test showed consistent success. Once connected to the Blue Pill, the system began providing expected output levels, confirming the issue was resolved.

The output was also tested using a simple code to turn the Blue Pill PC13 LED on and off whenever the signal value exceeded 600. Although the code provided consistent outputs in a stable environment, the movement of the electrodes caused flickering of the LED and the threshold needed to be changed whenever the muscle group was changed, so a dynamic threshold needed to be implemented.

4.3 Implementation of Adaptive EMG Control Logic

To improve the way the data is handled, and threshold is decided, two methods were used to reach the same target. The target was to find the average of a number of inputs at the start while the user has his muscle relaxed, multiply this by a factor and use that as a threshold. A calibration phase was implemented in order to find a baseline. The first method used was calculating a Moving Average (MA) of a set of one hundred inputs, but it was still affected by noise and large fluctuations.

The second method was using EMA for one hundred inputs one by one and inserting them into an array, then calculating their MA. This method proved successful in smoothing the signal and lowering the effect of noise. Knowing that EMA factor α needed to be less than 0.5 in order to have an average less affected by fluctuations, an α of 0.2 was found, through trial and error, to have the best effect.

Upon further consideration, the MA was changed, and instead the median was used on the values in the array in order to remove outliers (extreme values) caused by muscle twitches and motion artifact that skewed the results. A bubble sort was simple enough to apply and get the sorted array in minimal time. The array size was also changed to five hundred in order to find a more stable signal using more values.

After the initial calibration, the system would run EMA on each input and create a new buffer consisting of a small number of samples in order to adapt to changing environment. After long contractions, the baseline would shift upward, causing the system to get stuck in the “On” state. To fix this, the baseline buffer was made conditional, only allowing smoothed signal below the threshold to be added.

Multiplying the baseline with a scaling factor wasn't very accurate as the signal has inherent inconsistent fluctuations, so this natural fluctuation range was calculated by calculating variance that would be multiplied to the scaling factor K. The value of K was tested in the range of 1 to 10, and 5 consistently provided a more reliable threshold.

Flickering LED was still an issue especially during phases where state is changed. To ensure stability, a hysteresis band was added to introduce a dead zone where the LED keeps its state.

Variance also proved too reactive. It would spike during noise bursts and drop too low during stable signals, to prevent this, the variance was constrained to $\pm 10\%$ of the previous value in order to prevent instability while still allowing slow adaptation.

Multiple fails safes were also added to combat problems as they arise, and serial monitor was used to monitor all value calculated so they can be observed as they change [figure 4.3].

Voltage: 0.825 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.824 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.834 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.843 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.843 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.859 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.871 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.923 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.981 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 1.042 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.093 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.162 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.211 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.291 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.339 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.403 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON

Figure 4.3: Voltage, Baseline, Threshold, Variance and LED monitored

By the end of development, the system could reliably detect binary states from muscle contraction with no ML training or user specific tuning and regardless of muscle group.

4.4 Performance Evaluation of EMG-Based Control Across Different Users

The EMG sensor was tested on a total of ten users, six males and four females. The participants were of the same age group (22 to 25) but of different body types and muscle sizes. All subjects were not amputee.

The evaluation consisted of a set of predefined gestures or actions: individual finger flexion of thumb, index, middle and ring and little finger moving together, fist closure and wrist flexion. The EMG results were indicated using a simple LED that lights up if the movement is detected by the sensor.

The electrodes were connected to specific muscle groups related to the movement the user is instructed to do. Finger flexion, fist closure and wrist flexion were detected using the flexor digitorum superficialis and flexor digitorum profundus muscle groups.

All users' hands were cleaned with alcohol before placing conductive gel and Ag/AgCl electrodes. The same electrodes and gel were used for all users to ensure fairness. The same muscle sensor was also used, and it was connected to the same microcontroller running the same code. The battery connected was checked every ten minutes to ensure it does not change in value. All users were tested under the same conditions.

After the five second calibration phase the users were instructed to do each gesture in order and then they were asked to do whichever gestures they preferred at any speed they wanted.

The test was conducted to see if the algorithm used will be able to identify signals across different users and convert them to a usable output.

4.5 Challenges During 3D Printing and Physical Assembly in Local Conditions

Initially, the prosthetic hand was intended to be manufactured using a lightweight metal such as aluminum, due to its excellent strength to weight ratio. However, the intricate geometry of the hand required a high degree of precision that could only be achieved using a 5-axis CNC milling machine. After extensive research and outreach, it became evident that such equipment was not available locally in Egypt, and international outsourcing was not viable due to budgetary and time constraints.

As a result, two alternative fabrication methods were considered: sand casting and 3D additive manufacturing. Sand casting was initially explored as a low-cost method for producing metal components, but achieving the required dimensional accuracy and surface finish proved prohibitively expensive when offered by high-quality local factories. Additionally, the need for custom molds made it unsuitable for rapid prototyping.

Given these limitations, 3D printing emerged as the most accessible and practical solution, offering a flexible and low-cost method for producing complex geometries. It also allowed for quick reprinting of parts with any flaws. While not as mechanically robust as aluminum, high-quality Fused Deposition Modeling (FDM) prints using Polylactic Acid (PLA) was sufficient for initial testing and functionality validation.

The printer used had 0.1mm tolerance which produced dimensionally accurate components. The parts had an infill of 25% which provided enough structural integrity to withstand the typical forces from grasping and finger flexion without compromising the lightweight nature of the hand.

Despite accounting for a 0.1 mm clearance tolerance during design, certain components particularly the press-fit revolute joints between phalanges suffered from dimensional inaccuracies and could not be inserted in place. As an immediate solution, a metal nail of similar dimensions was placed instead of the pin, and its end was lightly soldered to form a cap and prevent it from

slipping out [Figure 4.4]. While not ideal for long-term use, this workaround allowed mechanical testing and further development to proceed without reprinting affected components.



Figure 4.4: Nails used as pins

The same problem occurred with the housing of the servo, which required using a grinder kit with a cutting disk attached to enlarge the housing.

The rubber parts that are used for cushioning were fabricated from Thermoplastic Polyurethane (TPU) due to its flexible rubber-like nature and damping characteristics. However, despite its advantage, its dimensional accuracy when 3D printed was significantly low. As a result, all 48 cushioning elements required manual trimming using precision cutters.

Despite being printed at one of the most advanced 3D printing facilities in Egypt, many parts were delivered with noticeable deformation or warping, which compromised their dimensional accuracy and fit [Figure 4.5]. This necessitated multiple rounds of reprinting to obtain parts that met the required specifications and tolerances for proper assembly.



Figure 4.5: Middle phalanx warped joint

Once the parts were confirmed to be intact and dimensionally accurate, the assembly of the 3D printed bionic was not difficult and was finished in under an hour.

4.6 Assembly and Wiring of the Complete Prosthetic Hand

Insertion of the strings into their housing proved to be a difficult task because of how flexible the strings were. This problem was quickly solved by submerging the end of the string in glue and holding it in an extended straight-line position using pliers. This made the wire rigid while still not increasing its diameter significantly making it very easy to insert in the housing. The solidified part is then cut, and the wire can be used.

The tendon wires were secured by tying one end around the motor pulleys, while the other end was fixed in its designated location within the prosthetic hand. Due to some wire fixation points being inaccurately printed, alternative attachment methods were employed. For instance, one wire was tied around a 3 mm nail, which was then firmly inserted into the palm structure to provide a secure anchor point.

The required ball bearing size of 2 mm was not found but an alternative was found by disassembling a Tilt Switch Module KY020 and using the metal sphere inside that is used to indicate tilting.

Two major challenges arose with the electrical components during assembly. First, soldering wires and pins securely in the confined spaces of the prosthetic hand proved very difficult, especially since PLA, the main housing material, is highly sensitive to heat and prone to deformation. Second, the originally designed wire channels and housings were too small, which required the wires to be placed in any gaps to be found and tied securely using tape.

To reduce the number of individual connections, the system ground and Vcc lines were soldered onto a piece of copper bus bar, allowing all other wires to connect more easily and efficiently.

The last problem faced was the dimensions of the battery presented by the seller were not accurate in thickness which caused the battery to protrude outside its housing causing the cover to not close properly.

4.7 Bionic Functionality and Load-Carrying Capacity

The prosthetic performed as expected with individual finger movement controlled by each motor. With adjustment to the rubber band, the fingers would reliably return to their original position after flexing.

One problem faced was that the pulleys on the motor shafts needed to be larger. Their small size caused the string to slip off and rotate around the shaft itself, which affected finger movement as the string was not getting the full rotation it should.

On one occasion, a motor wire disconnected, and the motor kept rotating continuously, causing a tendon path to break. This was solved by tightly wrapping extra wire around the broken part and adding a drop of glue, which worked perfectly to restore function.

Regarding the load-carrying capacity, the prosthetic hand showed promising strength. Unfortunately, the battery's extra thickness prevented the hand's cover from closing fully, making it difficult to grip the hand securely at its edges during testing. Regardless, the hand was tested with a 5 kg dumbbell while the whole system was turned off and the motors were free to rotate, meaning there was no holding torque, and it held the weight without any issues [figure 4.6].

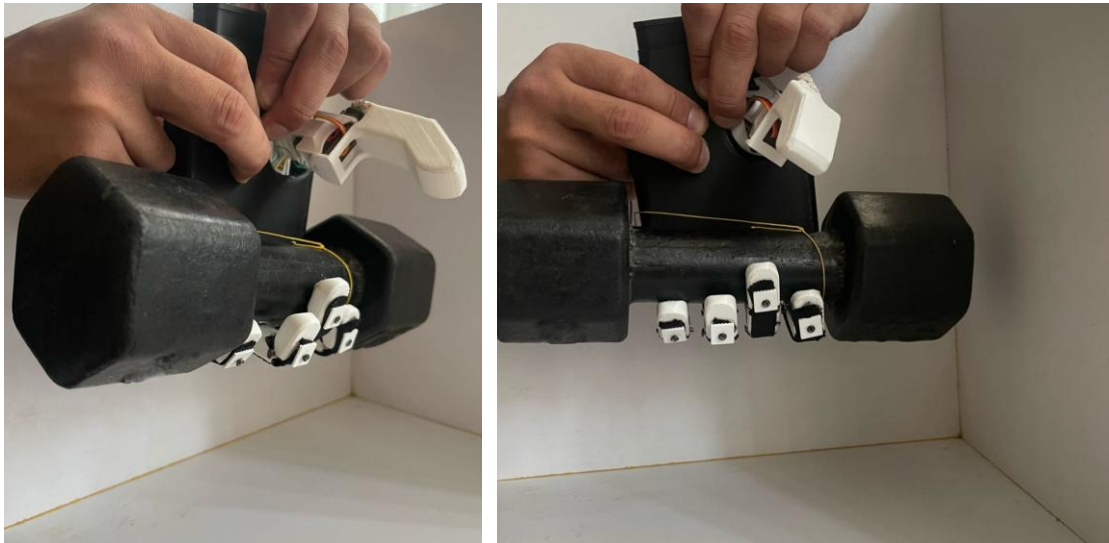


Figure 4.6: Bionic carrying 5Kg dumbbell

When the motors are activated, the high torque provided by the chosen motors should allow carrying significantly heavier loads. However, testing this precisely was difficult because of the incomplete assembly caused by the battery clearance issue.

An interesting observation is that the motors' high torque is both an advantage and a drawback: while it enables the hand to lift heavy objects, it also means the fingers cannot be moved manually when the motors are powered off, which could limit user interaction or emergency manual positioning.

Chapter 5

Results and Discussion

Multiple systems were tested to ensure functionality and determine what needs to be addressed in future work.

The 2S lipo battery charging system was tested to ensure safety and user comfort. The fully discharged battery was fully charged in 27 minutes using any charger that can supply 9V and 3A. It was able to withstand charging using Lenovo USB-C 150W charger [figure 5.1] with the battery temperature reaching 39°C compared to the ambient of 31°C . The battery functioned as expected, reaching temperatures safe enough for both user and internal components.



Figure 5.1: Prosthetic charged using Lenovo charger

The battery was not tested with all motors in constant use, but time was measured while the hand was being tested for different movements, and it lasted for six hours which is not ideal but expected as the battery capacity is only 1000 mAh. Expected usage time for prosthetics should be at least 24 hours to last the user an entire day.

Response time of EMG algorithm signal detection was very difficult to measure using a stopwatch due to the slow human reaction time of 0.25s. As an alternative the hand was plugged in the COM port and the serial monitor was used with 10ms delay between each reading and it showed almost instantaneous response to hand flexion.

Problems were faced during the placement of electrodes, which provided a steady output of 3.028V [figure 3.2]. Flexion of the muscle before placing the electrodes proved to be an effective way to solve the problem.

Voltage: 3.028 V	Baseline: 2.828 V	Threshold: 3.006 V	Variance: 0.001268	LED: OFF
Voltage: 3.028 V	Baseline: 2.828 V	Threshold: 3.006 V	Variance: 0.001268	LED: OFF
Voltage: 3.028 V	Baseline: 2.828 V	Threshold: 3.006 V	Variance: 0.001268	LED: OFF
Voltage: 3.028 V	Baseline: 2.828 V	Threshold: 3.006 V	Variance: 0.001268	LED: OFF
Voltage: 3.028 V	Baseline: 2.828 V	Threshold: 3.006 V	Variance: 0.001268	LED: OFF
Voltage: 3.028 V	Baseline: 2.828 V	Threshold: 3.006 V	Variance: 0.001268	LED: OFF

Figure 5.2: signal output showing constant 3.028 V

The algorithm was tested on ten different subjects and the electrodes were placed on different muscle groups and the users were instructed to do specific grips and movements each done ten times each. The flexor digitorum superficialis consistently gave the best reading for grip and wrist flexion with 100% accuracy in grip and wrist flexion, and 98.25% accuracy in individual finger flexion. From the four hundred finger movements, only seven mistakes were made by the algorithm.

The flexor digitorum profundus gave 100% accuracy in grip and wrist flexion, 78.75% accuracy in finger flexion. This value was significantly decreased due to the flexion of the ring and little finger flexion rarely being detected.

The algorithm proved very effective in detecting signals with very high accuracy and speed. Especially for grip and wrist flexion. For individual finger flexion, the flexor digitorum superficialis gave very high accuracy while the flexor digitorum profundus did not give satisfactory results, but that was because of the sEMG not outputting reading for the algorithm.

The algorithm was able to successfully calibrate itself regardless of user and muscle group. The threshold successfully adapted during the calibration phase and during any change in position by users. Figure 3.3 shows multiple serial monitor reading of different users and muscle groups with the threshold adjusted automatically by the algorithm.

Voltage: 0.303 V	Baseline: 0.277 V	Threshold: 0.463 V	Variance: 0.001380	LED: OFF
Voltage: 0.256 V	Baseline: 0.267 V	Threshold: 0.441 V	Variance: 0.001198	LED: OFF
Voltage: 0.282 V	Baseline: 0.268 V	Threshold: 0.431 V	Variance: 0.001067	LED: OFF
Voltage: 0.497 V	Baseline: 0.268 V	Threshold: 0.431 V	Variance: 0.001067	LED: ON
Voltage: 0.537 V	Baseline: 0.268 V	Threshold: 0.431 V	Variance: 0.001067	LED: ON
Voltage: 0.667 V	Baseline: 0.268 V	Threshold: 0.431 V	Variance: 0.001067	LED: ON

Voltage: 0.871 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.923 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 0.981 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: OFF
Voltage: 1.042 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.093 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON
Voltage: 1.162 V	Baseline: 0.807 V	Threshold: 0.977 V	Variance: 0.001155	LED: ON

Voltage: 0.643 V	Baseline: 0.551 V	Threshold: 0.709 V	Variance: 0.001000	LED: OFF
Voltage: 0.642 V	Baseline: 0.551 V	Threshold: 0.709 V	Variance: 0.001000	LED: OFF
Voltage: 0.644 V	Baseline: 0.551 V	Threshold: 0.709 V	Variance: 0.001000	LED: OFF
Voltage: 0.808 V	Baseline: 0.551 V	Threshold: 0.709 V	Variance: 0.001000	LED: ON
Voltage: 1.286 V	Baseline: 0.551 V	Threshold: 0.709 V	Variance: 0.001000	LED: ON
Voltage: 1.562 V	Baseline: 0.551 V	Threshold: 0.709 V	Variance: 0.001000	LED: ON

Voltage: 2.195 V	Baseline: 2.024 V	Threshold: 2.204 V	Variance: 0.001307	LED: OFF
Voltage: 2.205 V	Baseline: 2.024 V	Threshold: 2.204 V	Variance: 0.001307	LED: OFF
Voltage: 2.254 V	Baseline: 2.024 V	Threshold: 2.204 V	Variance: 0.001307	LED: OFF
Voltage: 2.347 V	Baseline: 2.024 V	Threshold: 2.204 V	Variance: 0.001307	LED: ON
Voltage: 2.361 V	Baseline: 2.024 V	Threshold: 2.204 V	Variance: 0.001307	LED: ON
Voltage: 2.263 V	Baseline: 2.024 V	Threshold: 2.204 V	Variance: 0.001307	LED: ON

Figure 5.3: serial monitor showing different thresholds of different users

Regarding the prosthetic, the weight carrying capacity of the bionic was shown in the experimental section to be high, with a 5Kg carrying capacity, without holding torque and was able to perform multiple grip patterns. The back rubber bands used to bring the fingers back sometimes didn't work as intended for all fingers and needed to be placed back in place manually every few iterations. The prosthetic was then tested for its ability to adapt and grip to different objects as shown in figures 5.4, 5.5 5.6 5.7 and 5.8.



Figure 5.4: holding a mug



Figure 5.5: Holding a phone



Figure 5.6: holding scissors



Figure 5.7: holding a pencil



Figure 5.8: Holding a filled water bottle

As shown in the previous figures, the hand was able to adapt to objects of different sizes and was even able to hold a filled 0.6L water bottle without slipping proving its effectiveness.

The hand was also able to perform multiple grip patterns, further proving its agility [figure 5.9].

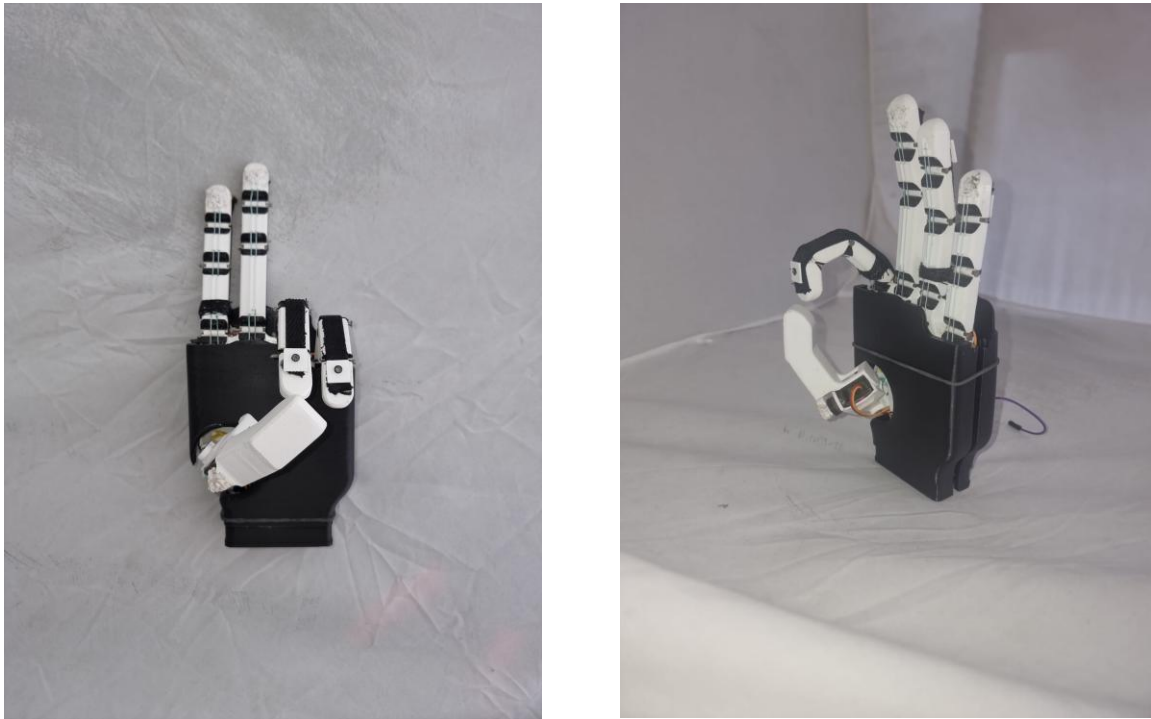


Figure 5.9: multiple grip patterns

Chapter 6

Conclusion and Future Work

This thesis successfully developed a low-cost, tendon-driven bionic hand controlled by sEMG signals, demonstrating significant potential for accessible prosthetic solutions in resource-constrained settings like Egypt. Costing 5000 EGP while enabling intuitive finger control and 5 kg load-carrying capacity, the bionic can be a starting prototype to be build and improved upon to create an accessible and high-quality open-source bionic.

The adaptive control algorithm, utilizing exponential moving average and dynamic thresholding, achieved 98.25% accuracy in detecting individual finger flexion and 100% accuracy for grip and wrist flexion across ten users. Despite variations in muscle signal strength the algorithm managed to adapt and maintain performance.

The use of 3D-printed PLA addressed cost and availability challenges, although limitations in manufacturing precision were encountered. The tendon-driven mechanism, enhanced by elastic bands and rubber damping, closely replicated the human hand, improving user experience.

This work provides a practical design methodology, mechanical assembly, signal processing, and control logic can be used to guide future low-cost prosthetic development.

Future improvements to the bionic hand include integrating pressure sensors on fingertips to provide haptic feedback, enabling finer control over grip force for objects of varying size and fragility.

Using sEMG can also be improved by developing a custom fabric sleeve with embedded electrodes to enhance comfort and reduce motion artifacts and upgrading to intramuscular electrodes, could greatly enhance signal accuracy and potentially restore sensory feedback through nerve rerouting.

Incorporating machine learning for pattern recognition could enable more complex grip patterns, though this would require balancing computational demands with cost.

Upgrading the battery system to support higher capacity (e.g., >1000 mAh) would extend operational time beyond six hours to avoid the hassle of charging the prosthetic multiple times a day.

Finally, conducting EMG tests with amputee users in real-world settings would validate long-term usability and identify further areas of improvement.

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